



US012387156B2

(12) **United States Patent**
Martin et al.

(10) **Patent No.:** **US 12,387,156 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **QUANTIFYING USAGE OF DISPARATE
COMPUTING RESOURCES AS A SINGLE
UNIT OF MEASURE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/079,359**

(22) Filed: **Oct. 23, 2020**

(65) **Prior Publication Data**

US 2021/0125128 A1 Apr. 29, 2021

Related U.S. Application Data

(60) Provisional application No. 62/924,772, filed on Oct.
23, 2019.

(51) **Int. Cl.**
G06Q 10/06 (2023.01)
G06Q 10/0631 (2023.01)
G06Q 30/0283 (2023.01)

(52) **U.S. Cl.**
CPC ... **G06Q 10/06315** (2013.01); **G06Q 30/0283**
(2013.01)

(58) **Field of Classification Search**
CPC G06Q 10/00–90/00
See application file for complete search history.

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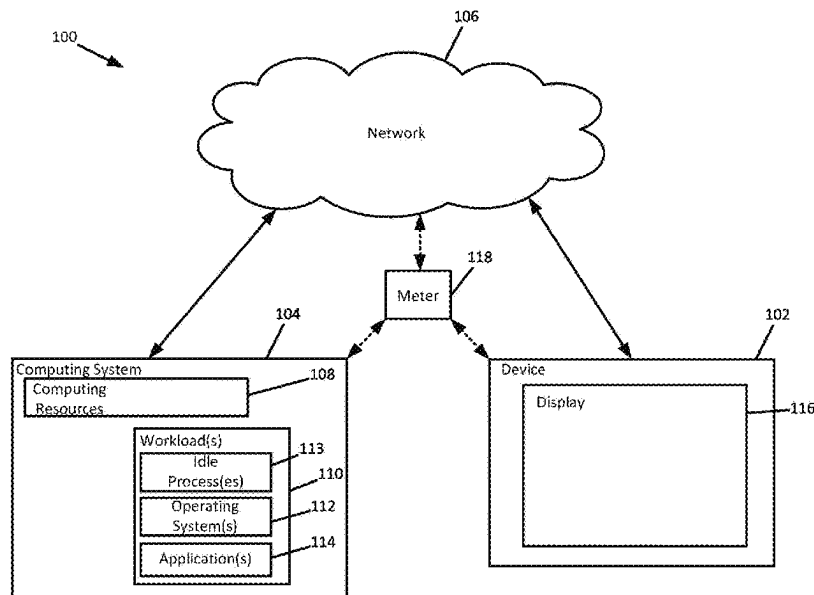
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(57) **ABSTRACT**

A method including: receiving a single usage value for a plurality of physical computing resources over a period of time; receiving a cost for the plurality of physical computing resources; and determining, using one or more processors, a normalized usage rate cost of the plurality of physical computing resources over the period of time by normalizing the cost for the plurality of physical computing resources for a unit of the single usage value.

16 Claims, 12 Drawing Sheets



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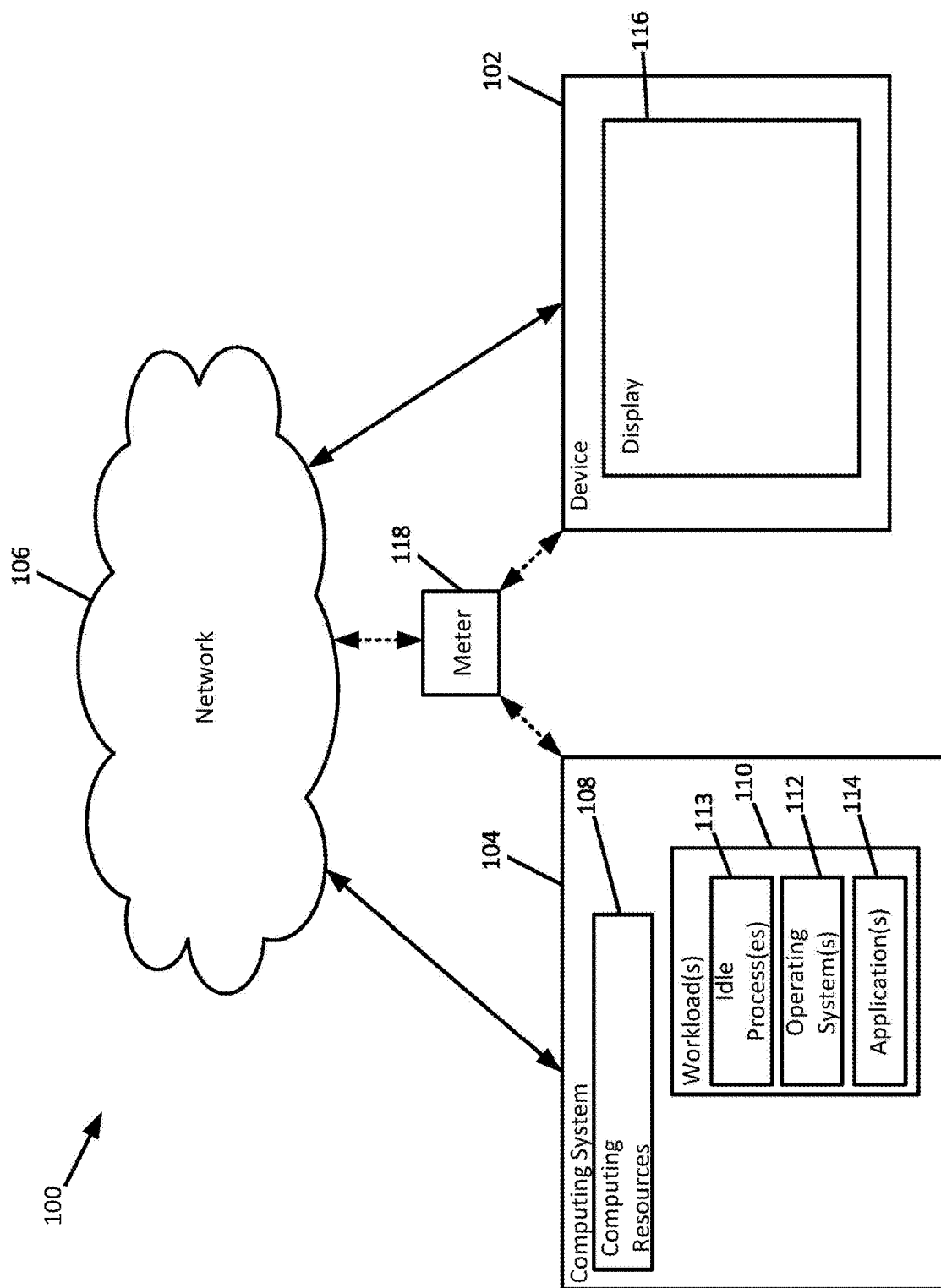


FIG. 1

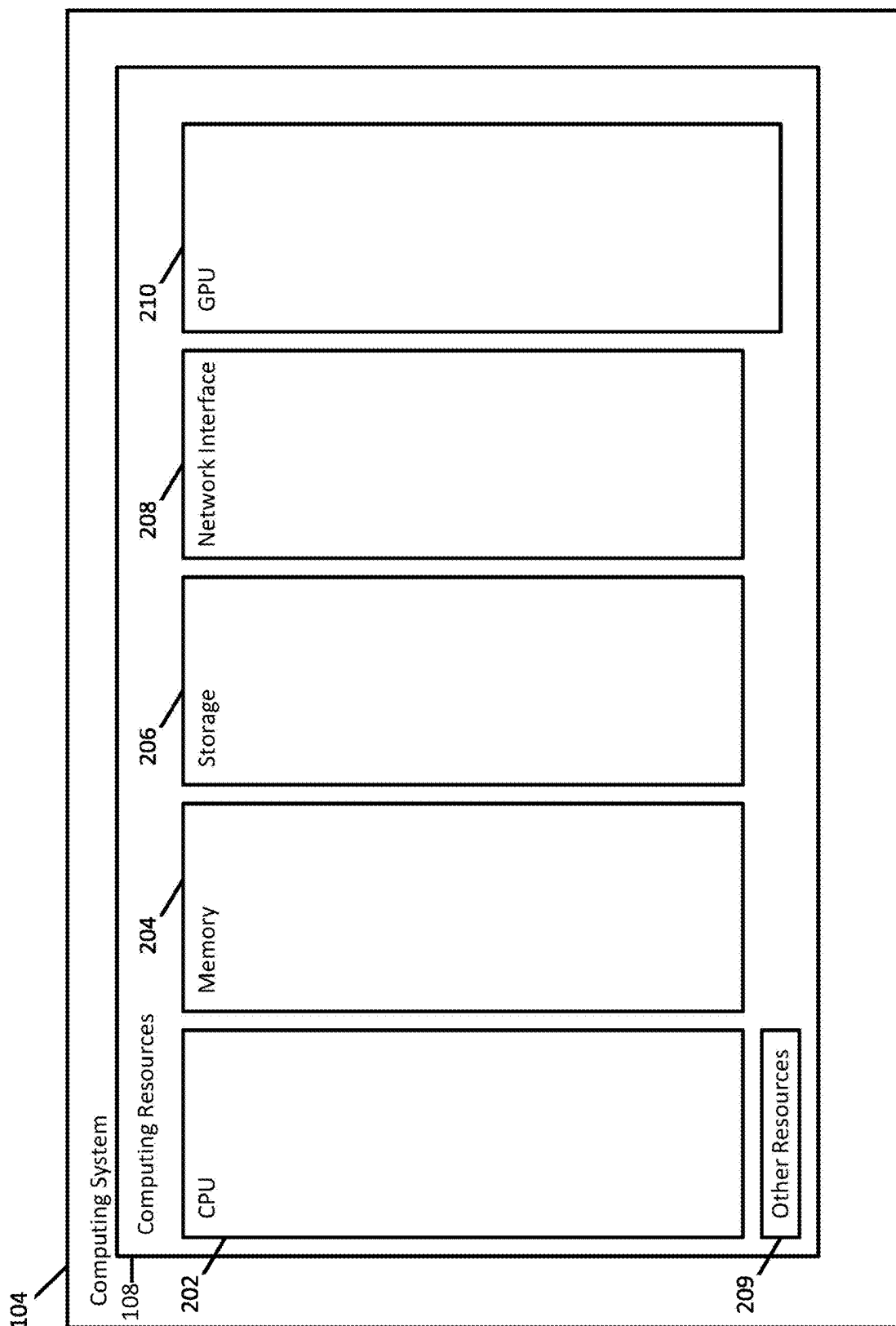


FIG. 2A

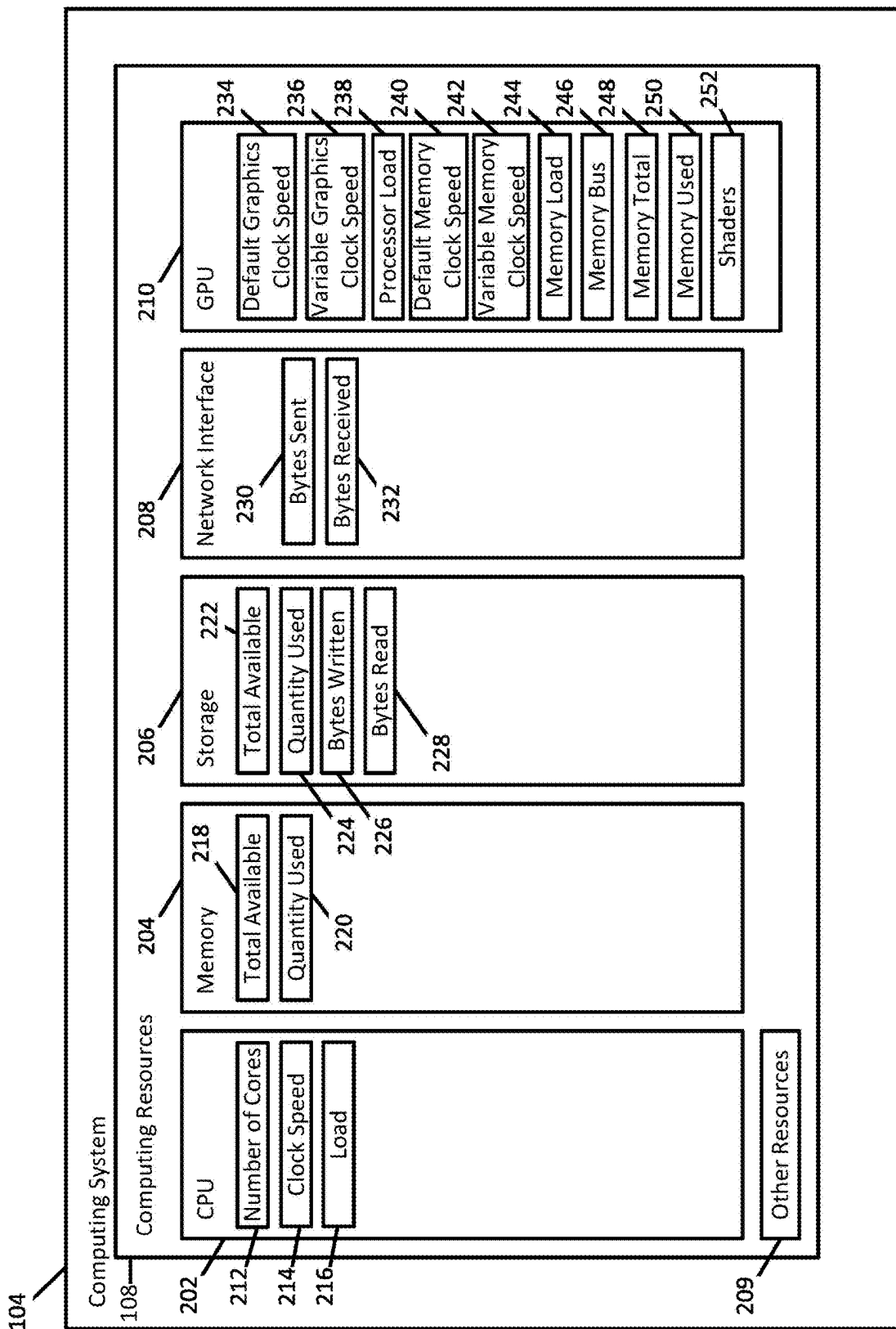


FIG. 2B

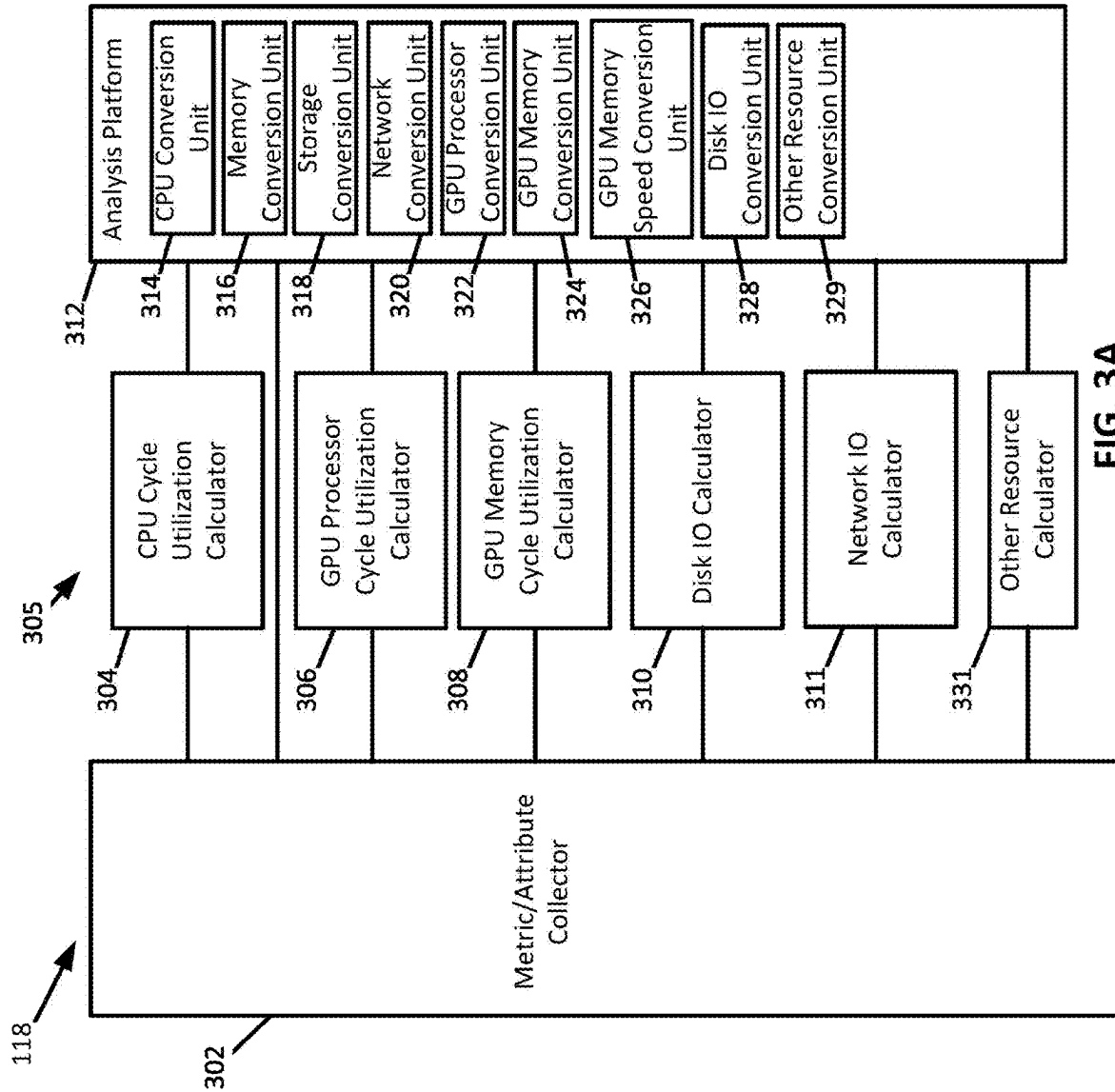
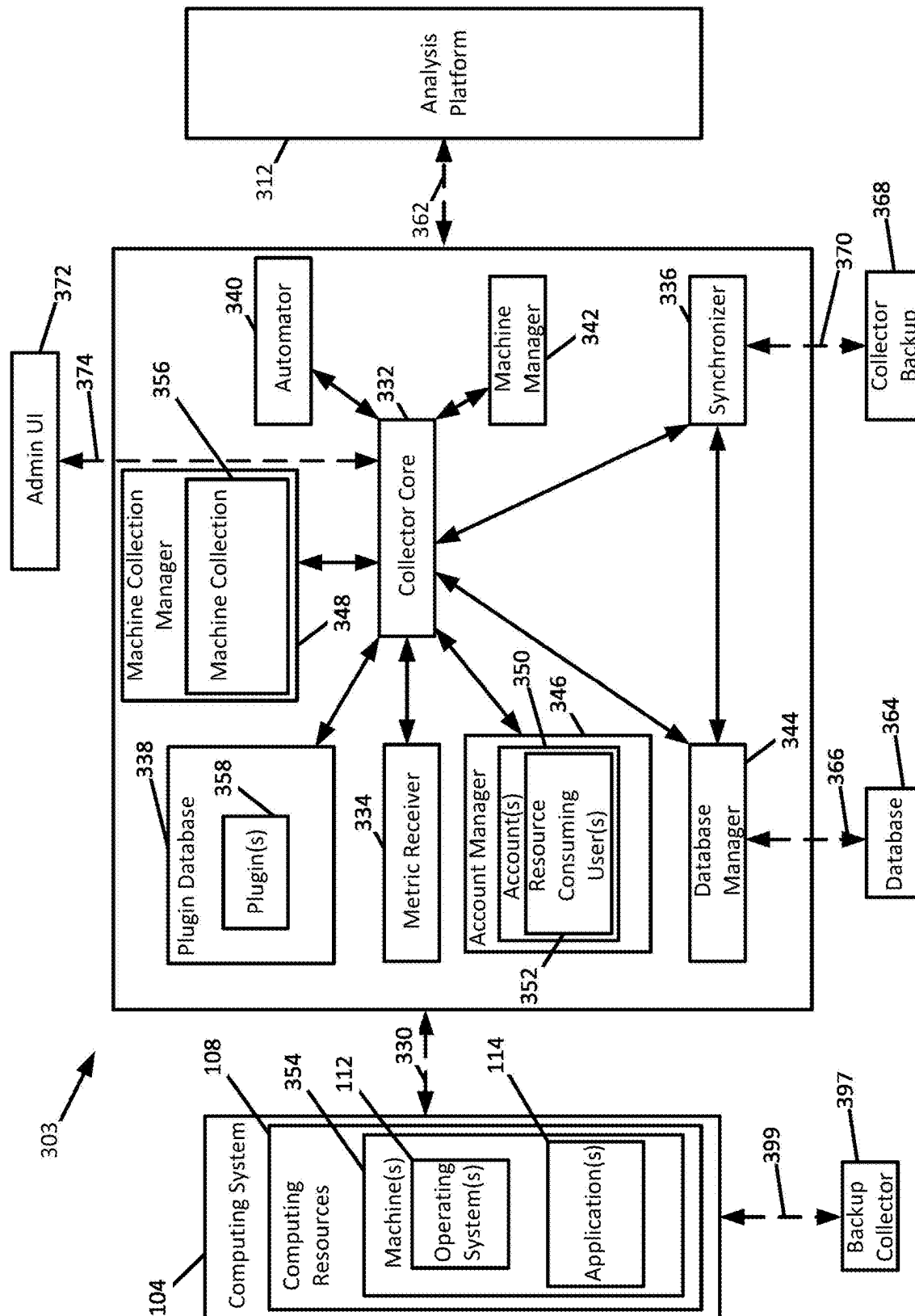


FIG. 3A



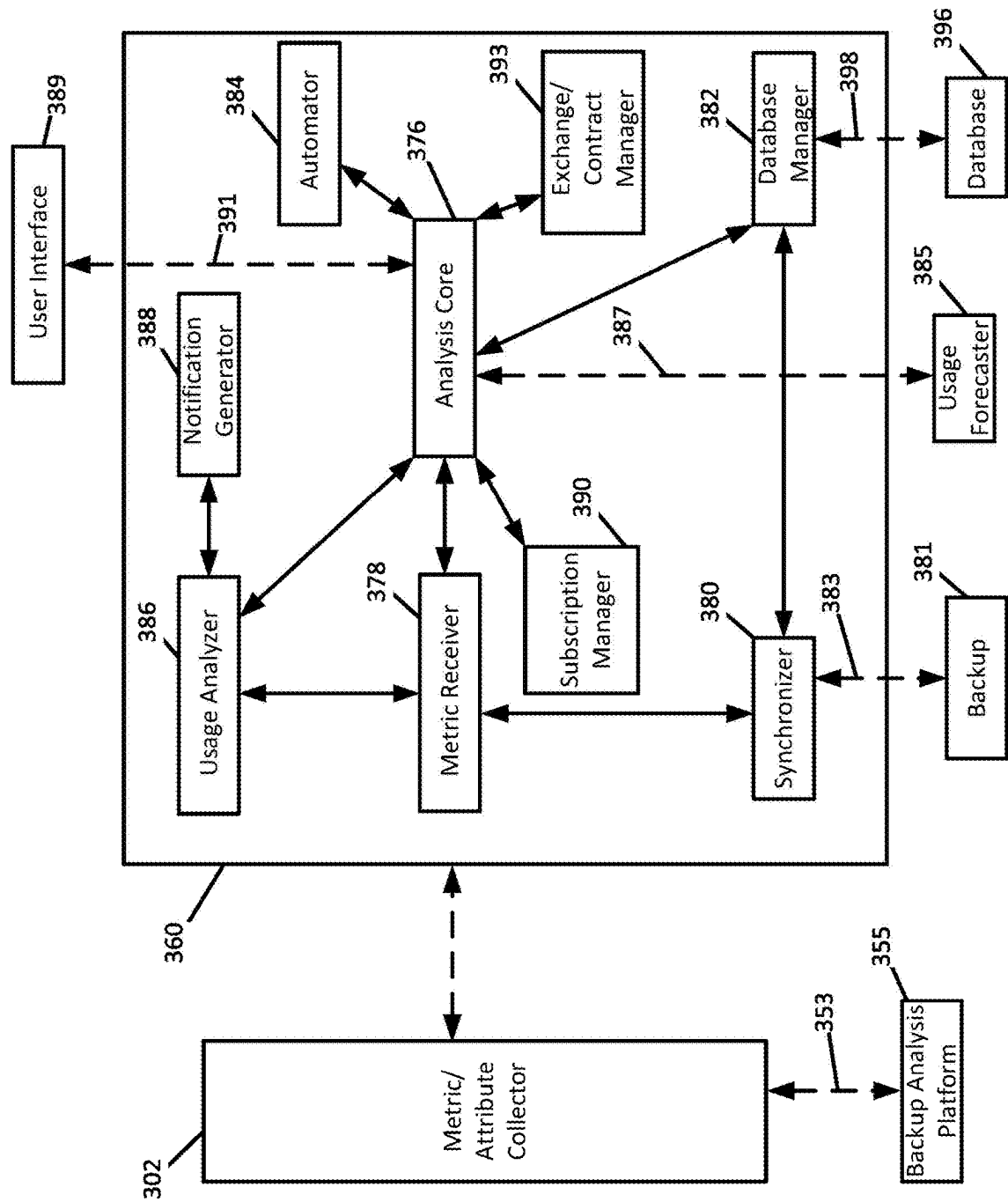


FIG. 3C

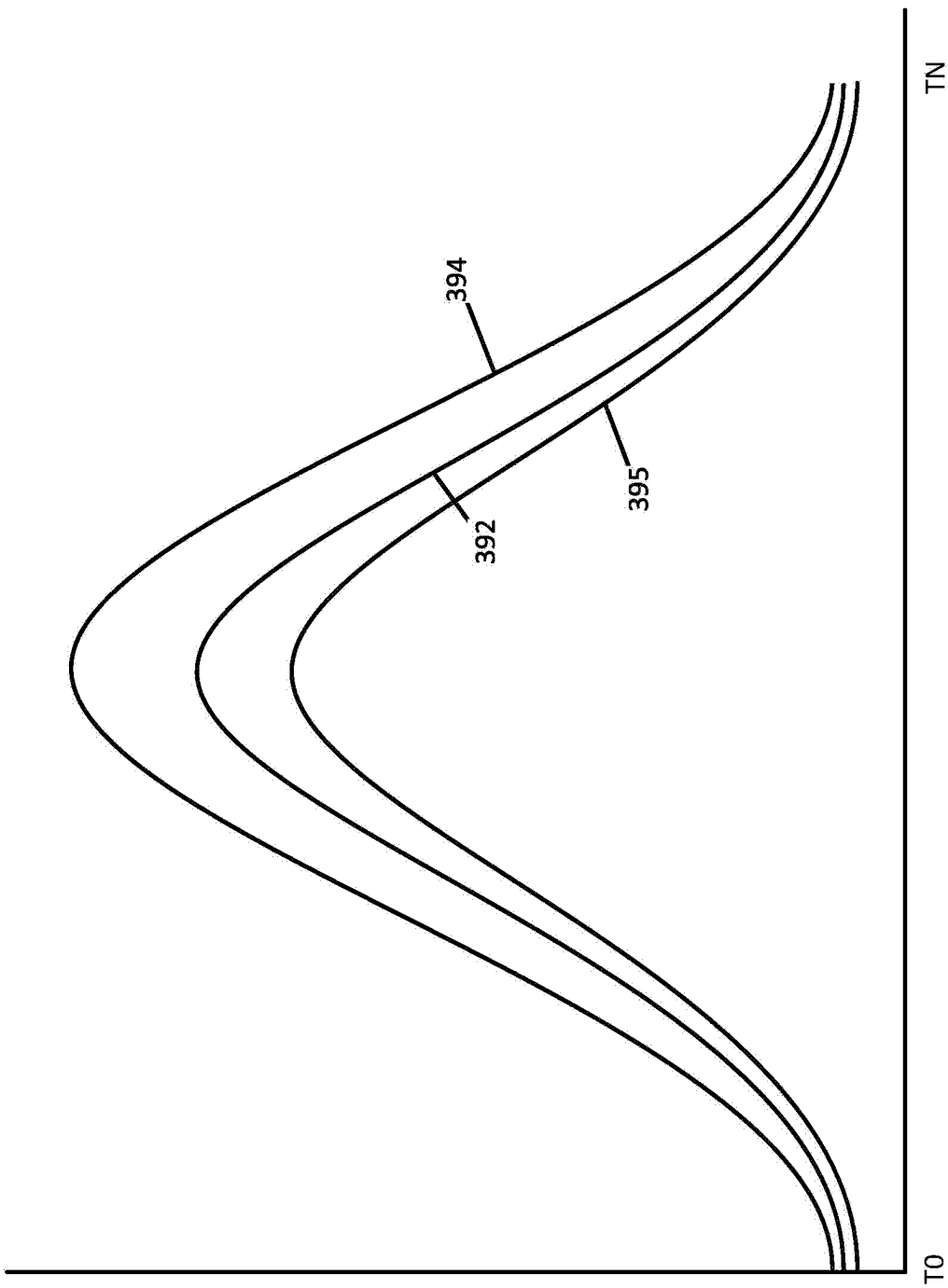


FIG. 3D

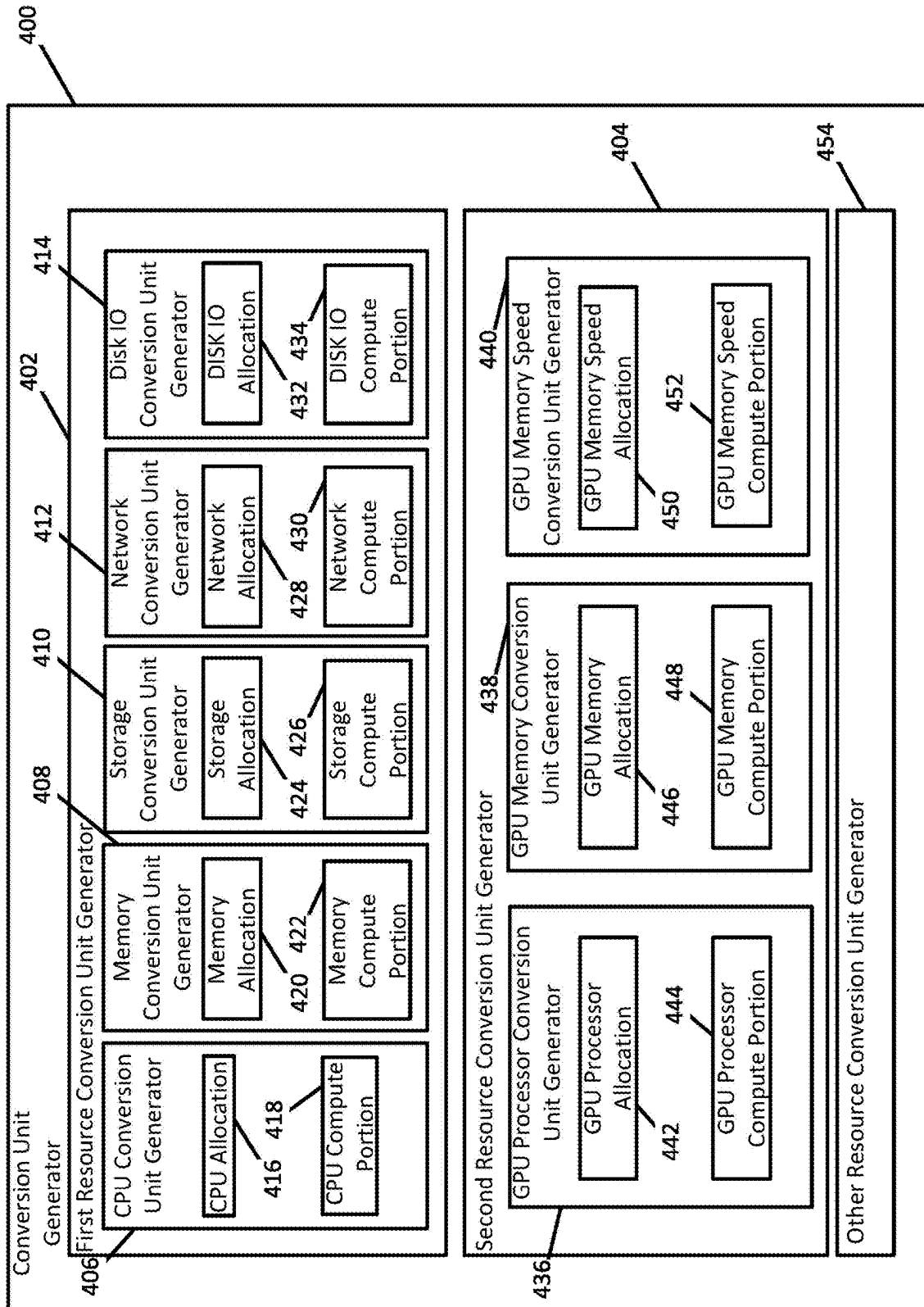
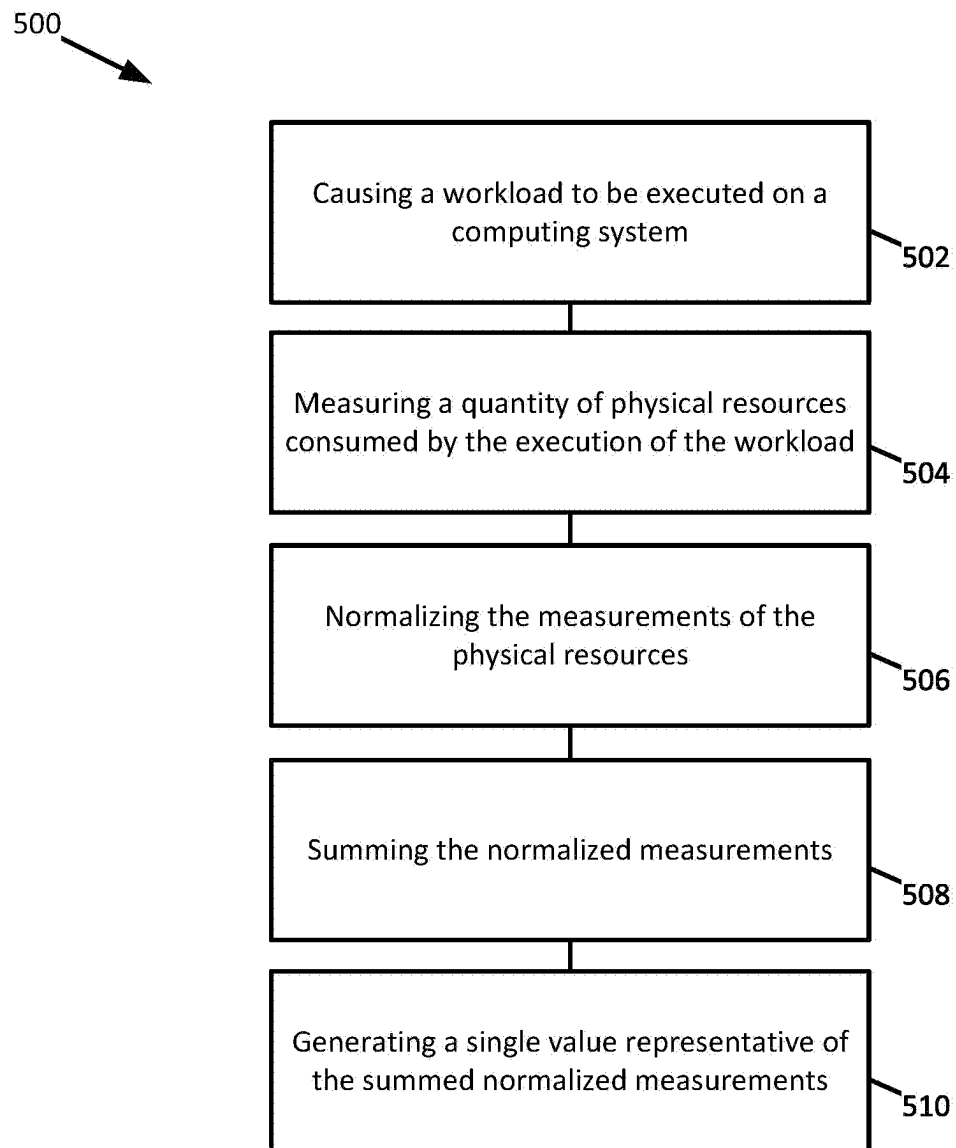
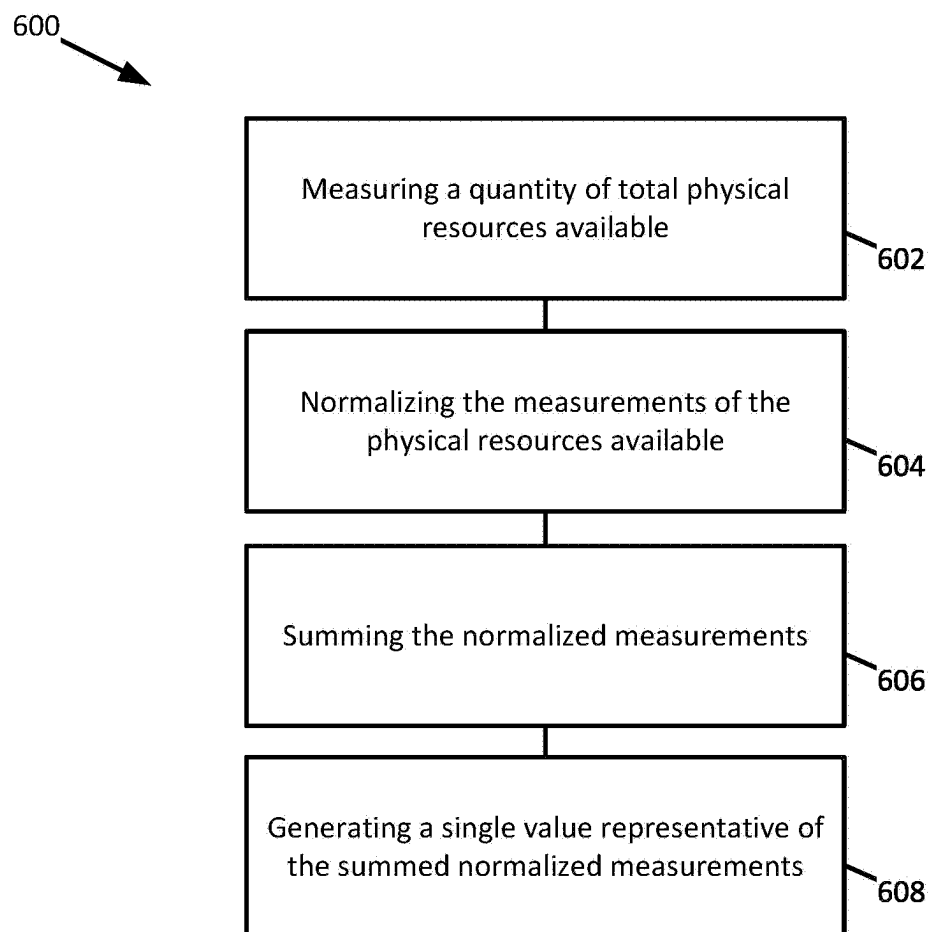
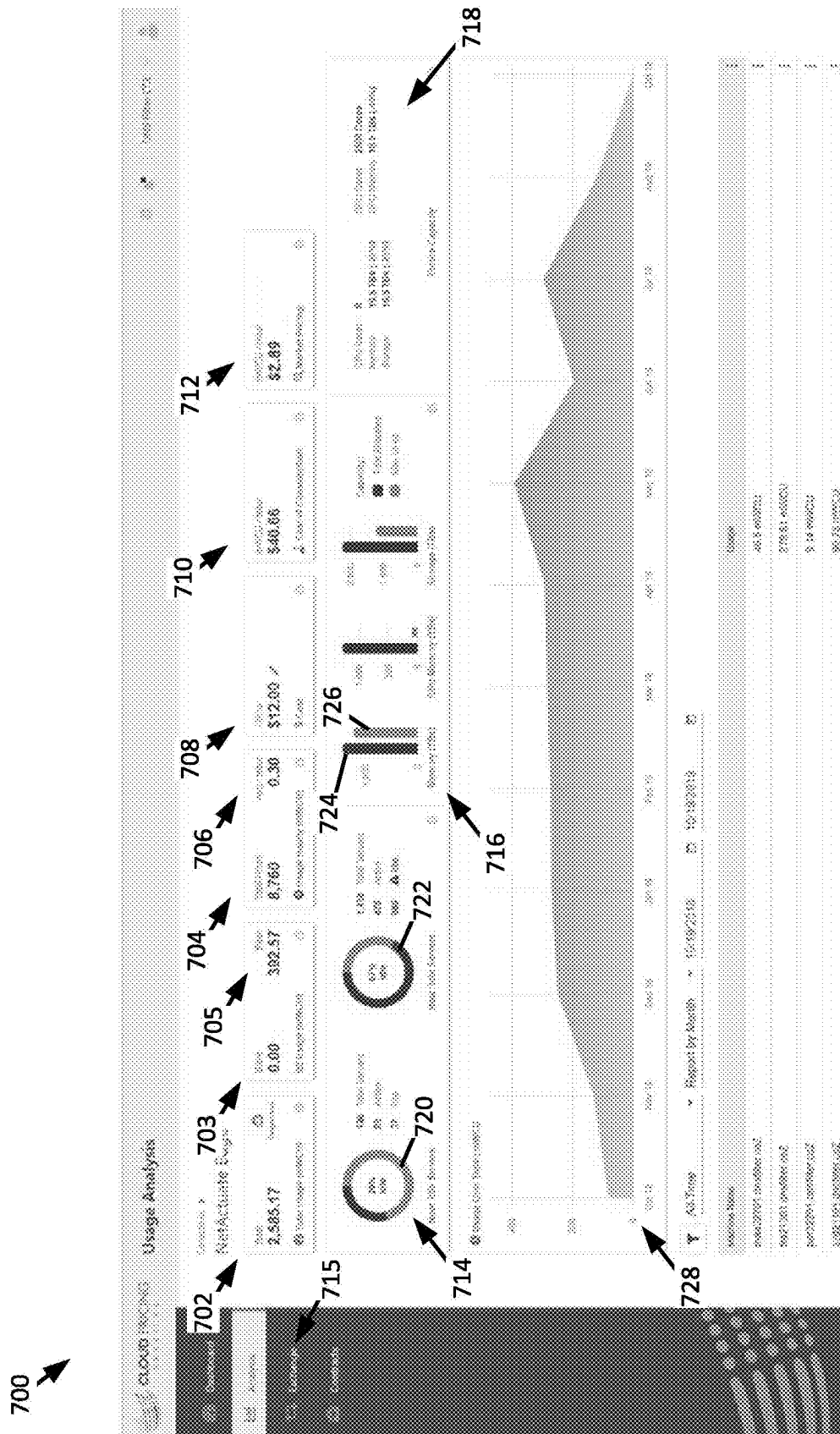


FIG. 4

**FIG. 5**

**FIG. 6**



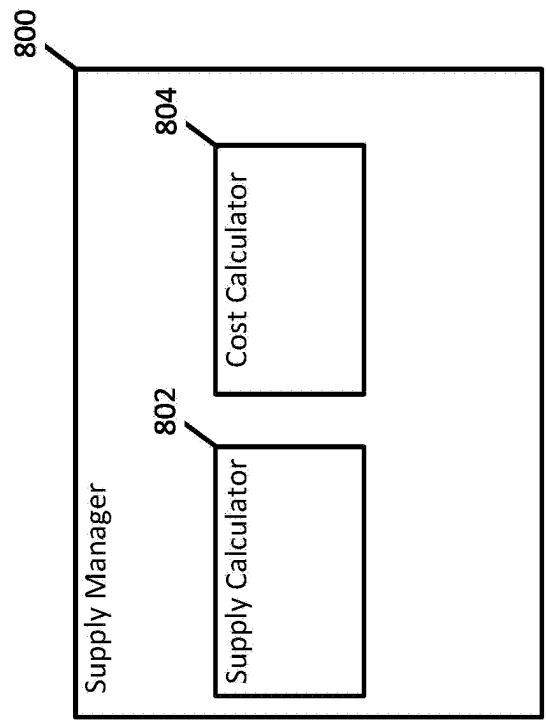


FIG. 8

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QUANTIFYING USAGE OF DISPARATE COMPUTING RESOURCES AS A SINGLE UNIT OF MEASURE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims the benefit of U.S. Provisional Application Ser. No. 62/924,772, filed on Oct. 23, 2019, the entirety of which is fully incorporated herein by reference.

TECHNICAL FIELD

The present disclosure is generally related to computing and more particularly to metering usage of disparate computer resources and quantifying the usage as a single unit of measure.

BACKGROUND

Cloud computing allows a user (e.g., an individual or an organization) to rent computing resources that are maintained and/or managed by a third-party. This may allow the user to rent computing resources that are tailored to their individual computing needs without necessarily having to make a substantial investment in physical hardware, data center real estate, electricity, data center personnel, and/or software licenses.

An example cloud computing system may include physical resources (e.g., a processor and memory) and/or virtualized resources (e.g., a virtualized processor and virtualized memory). A user accessing the cloud computing system may rent a quantity of resources capable of running their workloads. Cloud providers may offer their computing resources either as a virtual configuration or as a physical configuration. Oftentimes, the cloud provider will run a virtualization software on top of their underlying physical resources to share resources across different workloads and/or customers in an attempt to maximize their financial yield.

In some cases, users may select a similarly configured virtual server over a physical server when it is priced at a lower cost per unit of time (e.g., \$ per hr.) because it may be perceived to be less costly. However, because virtualization enables computing resources to be shared across different workloads and/or customers, performance may also be lesser for the virtual server in comparison with the physical configuration. In other words, the virtual server may not be able to perform as much work over a period of time for a user as the physical server because the underlying resources are being shared. In some cases, when costs of similar virtual and physical server configurations are compared as a normalized unit of work, the physical server may prove less costly than the virtual server.

SUMMARY

According to some embodiments, there is provided a method including: receiving a single usage value for a plurality of physical computing resources over a period of time; receiving a cost for the plurality of physical computing resources; and determining, using one or more processors, a normalized usage rate cost of the plurality of physical computing resources over the period of time by normalizing the cost for the plurality of physical computing resources for a unit of the single usage value.

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According to some embodiments, there is provided a method including: receiving a plurality of normalized costs for a unit of a single usage value for a plurality of computing environments; and determining a normalized cost index for a unit of the single usage value by combining the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features and advantages will be better understood by reading the following detailed description, taken together with the drawings, wherein:

FIG. 1 shows a schematic example of a computing network, consistent with embodiments of the present disclosure.

FIG. 2A shows a schematic example of a computing system capable of being used in the computing network of FIG. 1, consistent with embodiments of the present disclosure.

FIG. 2B shows a schematic example of the computing system of FIG. 2A, consistent with embodiments of the present disclosure.

FIG. 3A shows a schematic example of a meter for measuring usage of computing resources capable of being used in the computing system of FIG. 2, consistent with embodiments of the present disclosure.

FIG. 3B shows a schematic example of a metric/attribute collector capable of being used with the meter of FIG. 3A, consistent with embodiments of the present disclosure.

FIG. 3C shows a schematic example of an analysis platform capable of being used with the meter of FIG. 3A, consistent with embodiments of the present disclosure.

FIG. 3D shows a plot (or graph) illustrating resource usage, consistent with embodiments of the present disclosure.

FIG. 4 shows a schematic example of a conversion unit generator, which may be used with the meter of FIG. 3, consistent with embodiments of the present disclosure.

FIG. 5 shows an example flow chart of a method of quantifying usage of disparate computing resources as a single unit of measure, consistent with embodiments of the present disclosure.

FIG. 6 shows an example flow chart of a method of quantifying a total of disparate computing resources available (e.g., both used and inactive resources) as a single unit of measure, consistent with embodiments of the present disclosure.

FIG. 7 shows an example of a meter portal, consistent with embodiments of the present disclosure.

FIG. 8 shows a schematic example of a supply manager, consistent with embodiments of the present disclosure.

DETAILED DESCRIPTION

A system, method, and apparatus for measuring physical resource usage in computing and quantifying the usage as a single unit of measure is generally disclosed herein. For example, disparate computing resources may be measured, normalized (e.g., converted) to a common unit of measure, and each of the normalized measurements may be summed such that the total usage may be represented as a single value. By representing disparate computing resources as a single unit, resource usage (or consumption) can be monitored in real-time, workloads can be optimized to improve usage of available resources, and/or waste can be reduced.

By measuring physical resources (as opposed to virtual resources) a more accurate measure of actual usage may be obtained. For example, due to a technique used known as ballooning, measuring usage based on virtual resources may overestimate the actual usage of physical resources. In other words, when measuring only virtual resources, the virtual resources may indicate that more resources are being consumed (or used) by a workload than what the workload is actually consuming in physical resources. In some cases, the quantity of virtual resources indicated as being consumed may exceed the physical resources available.

By measuring physical resources (as opposed to virtual resources) a more accurate measure of actual work by the underlying computing resources may be obtained. In some cases, cloud computing users may select a virtual resource configuration over a physical resource configuration because it is priced at a lower cost per unit of time (e.g., \$ per hr.). However, because virtualization software may be used to enable computing resources to be shared across different workloads (and sometimes customers) in order to increase a supplier's yield, performance may also be lesser in comparison with the physical configuration, which may be dedicated to a user. In other words, the virtual resource configuration may not be able to perform as much work as the physical resource configuration because resources are shared. In some cases, when costs are compared as a unit of work, the physical resource configuration may prove less costly than the virtual resource configuration.

A graphics processing unit (GPU), as used herein, may generally refer to a computing resource that includes at least a graphics processor and/or graphics memory and, in some cases, the GPU may generally be referred to as a graphics processing system (GPS).

FIG. 1 shows a schematic example of a computing network 100. As shown, the computing network 100 may include a device 102 and a computing system 104, each being communicatively coupled to a network 106 (e.g., the internet). The computing system 104 may include physical computing resources 108 configured to execute one or more workloads 110. The one or more workloads 110 may include, for example, one or more of one or more operating systems 112, one or more applications 114, and/or one or more idle processes 113. The idle processes 113 may be background processes related to the operation of the computing system 104 (e.g., as a result of the operation of hardware of the computing system 104) and may consume at least a portion of the physical computing resources 108 even though a request from, for example, the device 102 to execute the application 114 and/or the operating system 112 has not been received. In other words, even when the computing system 104 is idle, the computing system 104 is still consuming at least a portion of the physical computing resources 108 due to the operation of the physical hardware of a computing system 104 when powered on.

One or more of the workloads 110 may be caused to be executed in response to a request generated by the device 102 (e.g., the operating system(s) 112 and/or the application(s) 114). For example, the device 102 may request the execution of one or more applications 114 by communicating the request to the computing system 104 through the network 106. When the request is received, the computing system 104 may allocate at least a portion of the physical computing resources 108 to the execution of the application 114. In response to the application 114 being executed, the computing system 104 may transmit data back to the device 102. For example, the data transmitted back to the device 102 may be used to generate a graphical user

interface on a display 116 of the device 102. A user of the device 102 may then interact with the graphical user interface and cause the device 102 to transmit additional instructions to the computing system 104 that cause different portions of the application 114 to be executed on the computing system 104.

As shown, a meter 118 may be provided. The meter 118 may include any combination of hardware, software, and/or firmware configured to measure the usage of the physical computing resources 108 used by the workloads 110 (e.g., the idle processes 113, the operating system(s) 112, and/or the application(s) 114) over a period of time (e.g., a predetermined or non-predetermined period of time). For example, the meter 118 may be implemented as software configured to be stored on one or more memories (e.g., any type of tangible, non-transitory storage medium, and may include any one or more of a magnetic recording medium (e.g., a hard disk drive), optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic and static RAMs, erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), flash memories, magnetic or optical cards, or any type of storage media for storing electronic instructions) and executed by one or more processors (e.g., processors offered for sale by INTEL, Advanced Micro Devices, IBM, ARM, Oracle, and/or any other processor) to cause one or more operations to be carried out. By way of further example, the meter 118 may be implemented as circuitry (e.g., an application-specific integrated circuit).

The usage of the physical computing resources 108 may be measured using a host operating system (e.g., an operating system written in an assembly language), a guest operating system and/or a hypervisor executing on the computing system 104. For example, the meter 118 may include a plugin database having one or more plugins configured to enable measuring of resource usage of the host operating system via a hypervisor executing on the computing system 104. By way of further example, the meter 118 may be configured to measure resource usage directly from the physical computing resources 108 (e.g., using a host operating system). In some cases, the meter 118 may measure the physical computing resource usage across multiple different computing systems 104.

The meter 118 measures usage of the physical computing resources 108, normalizes each of the physical computing resources 108 to a common unit of measure, and combines normalized computing resources such that the combined (e.g., total) usage can be represented as a single unit. In other words, the meter 118 may be generally described as quantifying the usage of multiple disparate computing resources as a single value representative of the usage of the computing resources 108. For example, the meter 118 may be configured to measure the usage of multiple disparate computing resources over a period of time and generate a single usage value representative of the usage over the period of time.

In order to mitigate and/or otherwise prevent the effects of ballooning, the meter 118 measures the physical resource usage as opposed to, for example, virtual resource usage. As such, a physical resource usage of, for example, a virtual machine can be measured more accurately. In some cases, the meter 118 may also be configured to measure a maximum quantity of the physical computing resources 108 available to be used (e.g., used and inactive resources), normalize the measured quantity of each physical resource of the physical computing resources 108 to a common unit of measure, and then sum the normalized resources to obtain

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a maximum quantity of physical resources available represented as a single unit. By measuring a maximum quantity of resources available, a user may be able to optimize the quantity of physical computing resources **108** for executing the workload **110**. Additionally, or alternatively, the meter **118** may be configured to measure a minimum quantity of the physical computing resources **108** required for running the computing system **104** (e.g., only the resources used by the idle processes **113**), normalize the measured quantity of each physical resource of the physical computing resources **108** to a common unit of measure, and then sum the normalized resources to obtain a single unit representative of a minimum quantity of physical resources required to be used (e.g., when the computing system **104** is operating but not executing one or more operating systems **112** and/or applications **114**).

In some cases, the meter **118** may be configured to measure the usage of physical resources over a period of time (e.g., a billing period for the resource usage, a contractual term, an hour, a day, a week, and/or any other period of time) and generate a single usage value representative of the usage of the physical resources over the period of time. The measured usage may then be utilized to determine the appropriate quantity of resources to purchase for a period of time and/or be used to generate a bill based on actual usage. As a result, the resource consuming user may reduce the quantity of resources purchased and the resource provider may reduce the total quantity of resources offered for sale and/or allocate the unused resources to another user.

In some cases, the meter **118** may be configured to enable and/or disable the device's **102** access to the computing resources **108** of the computing system **104**. In other words, the meter **118** may be configured to selectively communicatively couple and/or decouple the device **102** to the computing system **104**. For example, the meter **118** may disable access, by the device **102**, to at least a portion of the computing resources **108** after expiration of a period of time (e.g., the lapsing of a contractual term), failure to satisfy a contractual term (e.g., failure to pay a bill, misuse of computing resources, and/or the like), and/or in response to resource usage, as measured over a period of time, exceeding or failing to exceed a predetermined quantity (or threshold). By way of further example, the meter **118** may enable access to the computing resources **108** at a predetermined time (e.g., when a contractual term begins).

In some cases, the meter **118** may be configured to selectively enable and disable access to at least a portion of the computing resources **108** based on availability. For example, the device **102** may have access to at least a portion of the computing resources **108** when another user is not using the computing resources **108**. However, when another user requests access to the computing resources **108**, a recall request may be transmitted to the meter **118** that causes the meter **118** to disable access, by the device **102**, to at least a portion of the computing resources **108** such that the other user can access the computing resources **108**. As a result, the meter **118** may generally be described as being configured to reduce a quantity of the computing resources **108** that are not being used to, for example, execute one or more applications **114** or operating systems **112**. In some cases, the meter **118** may be able to predict a future recall request and notify the user of a pending recall of their computing resources **108** in advance of another user requesting access to the computing resources **108** or if a resource supplier seeks to terminate access to the computing resources **108** for any reason (i.e., billing term expiration, re-configuration of resources, and/or for any other reason). By way of further of example, the

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meter **118** may be able to predict a future recall request based on a demand increase for a certain computing platform and/or due to certain timing during an hour, day, week, month, year (i.e., for seasonality), or any other time interval.

As shown, the meter **118** may be communicatively coupled to the device **102**, the computing system **104**, and/or the network **106**. In some cases, at least a portion of the meter **118** may be included in the device **102** and/or the computing system **104**. Additionally, or alternatively, at least a portion of the meter **118** may be included in a third-party device communicatively coupled to one or more of the device **102** and/or the computing system **104** (e.g., using the network **106**).

The device **102** may include, any one or more of, a personal computer, a tablet computer, a cellphone, a smart phone, a smart watch, a smart television, a fitness tracker, a smart scale (for weighing an object), a smart thermostat, a smart security monitoring system (e.g., indoor and/or outdoor camera system, doorbell, alarm system, lock system, lighting system, and/or any other smart security monitoring system), a smart display (e.g., a display configured to communicate with the network **106**), smart healthcare systems (e.g., ultrasound systems, molecular imaging systems, computerized tomography scan systems, x-ray systems, and/or any other smart healthcare systems), remote health monitoring devices, gaming devices (e.g., gaming consoles, hand-held gaming devices, virtual reality headsets, and/or any other gaming device), Internet of Things devices (e.g., autonomous vehicles, connected home appliances, drones, robots, and/or any other Internet of Things device), a server, and/or any other device capable of communicating with the computing system **104**.

FIG. 2A shows a schematic example of the computing system **104** of FIG. 1. As shown, the physical computing resources **108** of the computing system **104** may include a central processing unit (CPU) **202**, memory **204**, storage **206**, a network interface **208**, and a graphics processing unit (GPU) **210**. Each of the CPU **202**, the memory **204**, the storage **206**, the network interface **208**, and the GPU **210** have one or more metrics/attributes associated therewith.

Additionally, or alternatively, the computing resources **108** may include other resources **209**. The other resources **209** may include, for example, any one or more of a Field Programmable Gate Array (FPGA), a Tensor Processing Unit (TPU), an Intelligence Processing Unit (IPU), a Neural Processing Unit (NPU), a Vision Processing Unit (VPU), a Digital Signal Processor (DSP), an Accelerated Processing Unit (APU), an Application-Specific Integrated Circuit (ASIC), a System-on-Chip (SoC), a programmable SoC, an Application Specific Standard Parts (ASSP), an Adaptive Compute Acceleration Platform (ACAP), a microcontroller, and/or any other computing resource.

As shown in FIG. 2B, the CPU **202** may be associated with metrics/attributes for a number of processing cores **212**, a processor clock speed **214**, and/or a processor load **216**. The number of processing cores **212** indicates the number independent processing units (or cores) that are available to the CPU **202**. The processor clock speed **214** indicates the default clock speed for each of the cores of the CPU **202**. The processor load **216** indicates an amount of computation work being carried out by the CPU **202**. The CPU **202** may be any computer processor, including, for example, single and/or multicore processors capable of executing computer instructions. Examples of the CPU **202** may include processors offered for sale by INTEL, Advanced Micro Devices, IBM, ARM, Oracle, and/or any other processor.

The memory **204** may be associated with metrics/attributes for a total quantity of memory available **218** and/or a quantity of memory used **220**. The memory **204** may be tangible, non-transitory, memory. The memory **204** may be volatile memory. For example, memory **204** may be random access memory (RAM). Examples of RAM may include, for example, static RAM (SRAM), dynamic RAM (DRAM), fast page mode DRAM (FPM DRAM), extended data out DRAM (EDO DRAM), synchronous dynamic RAM (SDRAM), double data rate SDRAM (DDR SDRAM), RAMBUS dynamic RAM (RDRAM), and/or any other form of RAM.

The storage **206** may be associated with metrics/attributes for a total quantity of storage available **222**, a total quantity of storage used **224**, a number of bytes written **226** to storage **206**, and/or a number of bytes read **228** from storage **206**. The storage **206** may be non-volatile memory. For example, the storage **206** may include any type of tangible, non-transitory storage medium, and may include any one or more of a magnetic recording medium (e.g., a hard disk drive), optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic and static RAMs, erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), flash memories, magnetic or optical cards, or any type of storage media for storing electronic instructions.

The network interface **208** may be associated with metrics/attributes for a number of bytes sent **230** and/or a number of bytes received **232**. The network interface **208** may be configured to communicatively couple to the network **106** (FIG. 1). For example, the network interface **208** may be a network interface controller configured to communicatively couple to one or more of an Ethernet network, a wireless network, a fiber channel network, a fiber distributed data interface (FDDI) network, a copper distributed data interface (CDDI) network, and/or any other network.

The meter **118** itself may be configured to calculate and sum the number of network bytes sent over a period of time for one or more network interface controllers or servers. The meter **118** may also be configured to calculate and sum the number of network bytes received over a period of time for one or more network interface controllers or servers. The meter **118** may also be configured to calculate and sum both the number of network bytes sent with the number of network bytes received over a period of time for one or more network interface controllers or servers. In some cases, resource suppliers may charge customers depending on network bytes sent or received, or both, and may round network bytes received up or down, network bytes sent up or down, or the summed amount of network bytes sent and received up or down. In any case, this may create inaccuracies of actual network usage. These rounding discrepancies may impact what a user may be billed for, and in other scenarios, may create challenges when a user is trying to accurately right-size their networking equipment needs. In some cases, resource suppliers may also falsify or fail to log/track network sent/received data transfers altogether, and the meter **118** may provide an accurate record of usage. In some instances, the meter **118** may also be deployed to measure computing resources that have been previously running (i.e., powered on with some workload activity) and therefore certain network sent/received data transfer statistics may have a pre-existing usage amount that is greater than zero. In this scenario, the meter **118** may be able to determine an appropriate starting point (i.e., as if the network interface controller or server were starting from zero

network sent/received data transfer) in order to accurately measure network sent/received data transfer only while the meter **118** has been running. In another example where the meter **118** has already been running and a server has been rebooted or a network interface controller has been restarted, swapped, activated, or re-activated, network received and/or sent data transfer statistics may automatically reset to zero. In this case, the meter **118** may be able retain the existing network sent and/or received data transfer statistics as a starting point (or baseline) and then sum all new network sent and/or received data transfer metrics to provide data transfer readings(s) that accurately reflect before the reboot, plus any new data transfer statistics from after the reboot, new network interface addition, or network interface restart.

The GPU **210** may be associated with metrics/attributes for a graphics processor default clock speed **234**, a graphics processor variable clock speed **236**, a graphics processor load **238**, a graphics memory default clock speed **240**, a variable graphics memory clock speed **242**, a graphics memory load **244**, a graphics memory bus **246**, a total quantity of graphics memory available **248**, a quantity of graphics memory used **250**, and/or a quantity of shaders **252**. The graphics processor default clock speed **234** indicates the default clock speed for the processor of the GPU **210** and the graphics processor variable clock speed **236** indicates variable clock speed of the graphics processor. In some cases, the graphics processor variable clock speed **236** may exceed the clock speed of the graphics processor default clock speed **234**. In other cases, the graphics processor variable clock speed **236** may not exceed the graphics processor default clock speed **234**. Similarly, the graphics memory default clock speed **240** indicates the default clock speed of the graphics memory and the variable graphics memory clock speed **242** indicates the clock speed at which the graphics memory is actually running. In some cases, the variable graphics memory clock speed **242** may measure greater than the graphics memory default clock speed **240**. In other cases, the variable graphics memory clock speed **242** may measure less than (or equal to) the graphics memory default clock speed **240**. The graphics memory load **244** indicates a quantity of read and/or write operations being carried out on the GPU memory. The graphics memory bus **246** indicates a bus size of the GPU **210**.

The GPU **210** may include integrated and/or dedicated GPUs. For example, the GPU may include any of the GPUs made by, for example, INTEL, Nvidia, Advanced Micro Devices, and/or any other GPU. When the GPU **210** includes an integrated GPU, the metrics/attributes may be based, at least in part, on the shared resources of the integrated GPU.

While CPU **202**, memory **204**, storage **206**, network interface **208**, and GPU **210** may be referred to singularly, it should be understood that the computing system **104** may include multiple physical CPUs, memories, storages, network interfaces, and GPUs, one or more of which may be configured to cooperate to, for example, improve the performance of the computing system **104**. Similarly, in cases where the computing system **104** includes one or more of the other resources **209**, there may be one or more of each of the other resources **209**.

FIG. 3A shows a schematic example of the meter **118** of FIG. 1. As shown, the meter **118** may include a metric/attribute collector **302** communicatively coupled to the physical computing resources **108** (FIG. 2), calculators **305**, and an analysis platform **312**. Each of the metric/attribute collector **302**, the calculators **305**, and/or the analysis platform **312** may be distributed across different computing systems (or environments). In some cases, one or more of

the metric/attribute collector **302**, the calculators **305**, and/or the analysis platform **312** may be located on the same computing system. For example, one or more of the metric/attribute collector **302**, the calculators **305**, and/or the analysis platform **312** may be located on the device **102**. By way of further example, one or more of the metric/attribute collector **302**, the calculators **305**, and/or the analysis platform **312** may be located on a separate device communicatively coupled to the network **106** and/or the computing system **104**. By way of still further example, one or more of the metric/attribute collector **302**, the calculators **305**, and/or the analysis platform **312** may be located on the computing system **104**. In some cases, for example, the metric/attribute collector **302** may include the analysis platform **312** or the analysis platform **312** may include the metric/attribute collector **302**.

The metric/attribute collector **302** requests one or more metrics/attributes from one or more of the CPU **202**, the memory **204**, the storage **206**, the network interface **208**, the GPU **210**, and/or one or more of the other resources **209**. For example, the metric/attribute collector **302** can request one or more metrics/attributes from a hypervisor (e.g., using a plugin) executing on one or more of the CPU **202**, the memory **204**, the storage **206**, the network interface **208**, the GPU **210**, and/or one or more of the other resources **209**. By way of further example, the metric/attribute collector **302** can request one or more metrics/attributes from a host operating system (e.g., an operating system written in an assembly language) executing on one or more of the CPU **202**, the memory **204**, the storage **206**, the network interface **208**, the GPU **210**, and/or one or more of the other resources **209**. In some cases, the metric/attribute collector **302** may request system information corresponding to the computing system **104** and/or the computing resources **108**, cost and/or pricing information related to the usage of the computing system **104** and/or the computing resources **108**, hardware information relating the computing system **104** and/or the computing resources **108**, metadata relating to the computing system **104** and/or the computing resources **108** (e.g., server tag information, server instance names, and/or other metadata information), and/or the like. While the metric/attribute collector **302** may be configured to receive metrics/attributes in real-time (or near real-time), in some cases, the metric/attribute collector **302** may receive the metrics/attributes in the form of a data file such as a JavaScript Object Notation (JSON) file, a Comma Separated Values (CSV) file, a spreadsheet file, and/or any other file type. The received data file may be generated by a user, a third-party application (e.g., via a third-party application programming interface), and/or any other source. In some cases, a third-party application (e.g., third-part application programming interface) may provide metrics/attributes in real-time (or near real-time).

When the metric/attribute collector **302** requests one or more metrics/attributes from the CPU **202**, the requested metrics/attributes may be input into a CPU cycle utilization calculator **304**. The CPU cycle utilization calculator **304** may generate a metric/attribute corresponding to a quantity of CPU cycles being utilized based, at least in part, on the requested metrics/attributes from the CPU **202**. For example, the CPU cycle utilization calculator **304** may generate a CPU cycle utilization based, at least in part, on the number of processing cores **212**, the processor clock speed **214**, and the processor load **216**. In these cases, the CPU cycle utilization calculator **304** may generate the CPU cycles being utilized according to the following equation:

$$\frac{(\text{processor clock speed})}{\frac{(\text{number of processing cores})}{100}} * (\text{processor load}) \quad [\text{EQ1}]$$

When the metric/attribute collector **302** requests one or more metrics/attributes from the GPU **210**, the requested metrics/attributes may be input into one or more of a GPU processor cycle utilization calculator **306** and/or a GPU memory cycle utilization calculator **308**. The GPU processor cycle utilization calculator **306** and the GPU memory cycle utilization calculator **308** may generate a metric/attribute corresponding to a GPU processor cycle utilization and a GPU memory cycle utilization, respectively, based, at least in part, on the requested metrics from the GPU **210**.

For example, the GPU processor cycle utilization calculator **306** may generate a GPU processor cycle utilization based, at least in part, on the graphics processor variable clock speed **236** and the graphics processor load **238**. In these cases, the GPU processor cycle utilization calculator **306** may generate the GPU processor cycles utilized according to the following equation:

$$\left(\frac{\text{graphics processor variable clock speed}}{100} \right) * (\text{graphics processor speed}) \quad [\text{EQ2}]$$

By way of further example, the GPU memory cycle utilization calculator **308** may generate a GPU cycle utilization based, at least in part, on the variable graphics memory clock speed **242** and the graphics memory load **244**. In these cases, the GPU memory cycle utilization calculator **308** may generate the clock cycles of GPU memory consumed according to the following equation:

$$\left(\frac{\text{variable graphics memory clock speed}}{100} \right) * (\text{graphics memory load}) \quad [\text{EQ3}]$$

When the metric/attribute collector **302** requests one or more metrics/attributes from the storage **206**, the metrics/attributes may be input into a disk IO calculator **310**. The disk IO calculator **310** may generate a metric/attribute corresponding to a total number of bits read and written to the storage **206** over a period of time (e.g., one second). Therefore, in some cases, the disk IO calculator **310** may add the number of bytes written **226** and the number of bytes read **228** over the period of time to generate the metric/attribute corresponding to disk IO. In cases where the period of time is greater than one second, the number of bytes written **226** and the number of bytes read **228** may be divided by the total number of seconds. In cases where the period of time is less than one second, the number of bytes written **226** and the number of bytes read **228** may be divided by the fraction of a second.

When the metric/attribute collector **302** requests one or more metrics/attributes from the network interface **208**, the metrics/attributes may be input into a network IO calculator **311**. The network IO calculator **311** may generate a metric/attribute corresponding to a total number of bits transmitted or received by the network interface **208** over a period of time (e.g., one second). Therefore, in some cases, the network IO calculator **311** may add the number of bytes sent

230 and the number of bytes received 232 over the period of time to generate the metric/attribute corresponding to network IO. In cases where the period of time is greater than one second, the number of bytes sent 230 and the number of bytes received 232 may be divided by the total number of seconds. In cases where the period of time is less than one second, the number of bytes sent 230 and the number of bytes received 232 may be divided by the fraction of a second.

As shown, the analysis platform 312 may receive one or more of the metrics/attributes associated with the physical computing resources 108 including, for example, one or more metrics/attributes output from one or more of the calculators 305. In some cases, the analysis platform 312 may include one or more other resources calculators 331 that correspond to a respective one or more of the other resources 209. The one or more metrics/attributes output from one or more of the calculators 305 may include, for example, one or more of the CPU cycle utilization (e.g., as output from the CPU cycle utilization calculator 304), the GPU processor cycle utilization (e.g., as output by the GPU processor cycle utilization calculator 306), the GPU memory cycle utilization (e.g., as output by the GPU memory cycle utilization calculator 308), the disk IO (e.g., as output by the disk IO calculator 310), and/or network IO (e.g., as output by the network IO calculator 311). The analysis platform 312 may convert the received metrics/attributes using one or more conversion units. In some cases, each metric/attribute has a corresponding conversion unit. For example, the analysis platform 312 may utilize a CPU conversion unit 314, a memory conversion unit 316, a storage conversion unit 318, a network conversion unit 320, a GPU processor conversion unit 322, a GPU memory conversion unit 324, a GPU memory speed conversion unit 326, a disk IO conversion unit 328, and/or any other conversion unit corresponding to a computing resource (e.g., a conversion unit for one or more of the other resources 209 such as FPGA resources, ASIC resources, SoC resources, DSP resources, microcontroller resources, ACAP resources, and/or the like). While the analysis platform 312 may be configured to receive metrics/attributes in real time, in some cases, the analysis platform 312 may receive the metrics/attributes in the form of a data file such as a JavaScript Object Notation (JSON) file, a Comma Separated Values (CSV) file, a spreadsheet file, and/or any other file type. The received data file may be generated by a user, a third-party application (e.g., via a third-party application programming interface), and/or any other source.

The analysis platform 312 may convert the disparate computing resources into a normalized unit by dividing the usage of each physical resource (e.g., CPU 202, memory 204, storage 206, network interface 208, and GPU 210) by a respective conversion unit corresponding to the physical resource (e.g., the CPU conversion unit 314, the memory conversion unit 316, the storage conversion unit 318, the network conversion unit 320, the GPU processor conversion unit 322, the GPU memory conversion unit 324, the GPU memory speed conversion unit 326, and the disk IO conversion unit 328) to obtain a respective normalized usage for each physical resource. In some cases, the analysis platform 312 can be configured to convert one or more of the other resources 209 to a standardized unit. For example, the analysis platform 312 may include an other resource conversion unit 329.

The normalized usages can then be summed to obtain a total usage value. Additionally, or alternatively, as should be readily understood in view of the disclosure herein, in some

cases, the normalized usages can be summed according to a computing resource type (e.g., the CPU 202, the memory 204, the storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209) to obtain a normalized usage value for a specific resource. For example, a normalized usage of the physical resources of the GPU 210 (e.g., GPU processor and/or GPU memory) can be summed to generate a total GPU usage. Other examples for normalized usage of specific physical resources may include a total FPGA usage, a total ASIC usage, a total DSP Usage, a total SoC usage, a total microcontroller usage, a total ACAP usage, a total TPU usage, and/or any other total usage corresponding to a respective resource.

FIG. 3B shows a metric/attribute collector 303, which may be an example of the metric/attribute collector 302, that is configured to monitor resource usage of multiple (e.g., a plurality of) different users and associate each user's usage with a respective user. In other words, the metric/attribute collector 303 is configured to receive metrics corresponding to resource usage. The metric/attribute collector 303 may include any combination of hardware, software, and/or firmware.

The metric/attribute collector 303 may be configured to be included with any one or more of the computing system 104, the device 102, and/or any other device communicatively coupled to the network 106. For example, the device 102 may include the metric/attribute collector 303. When the device 102 includes the metric/attribute collector 303, a user of the metric/attribute collector 303 may be able to utilize the meter 118 with fewer modifications to the user's network security, when compared to when the metric/attribute collector 303 is not executed on the device 102.

As shown, the metric/attribute collector 303 is communicatively coupled to the computing system 104 via a communication link 330. The communication link 330 may transmit data (e.g., metrics/attributes associated with the computing resources 108) to the metric/attribute collector 303. Data transmitted over the communication link 330 may be encrypted to prevent unauthorized access to the data. As shown, the metric/attribute collector 303 may be remote from the computing system 104 (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the metric/attribute collector 303 may be local to the computing system 104 (e.g., operating on the same server, on the same network, and/or on the same device).

The metric/attribute collector 303 may also be configured to communicate with the analysis platform 312 over a communication link 362. Data transmitted over the communication link 362 may be encrypted to prevent unauthorized access to the data. In some cases, the metric/attribute collector 303 may include the analysis platform 312. As such, the metric/attribute collector 303 may not transmit data over the communication link 362.

The analysis platform 312 may generate the total resource usage value represented as a single normalized unit. As shown, the analysis platform 312 may be remote from the metric/attribute collector 303 (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the analysis platform 312 may be local to the metric/attribute collector 303 (e.g., operating on the same server, on the same network, and/or on the same device).

As shown, the metric/attribute collector 303 may include at least one collector core 332, at least one collector metric receiver 334, at least one collector synchronizer 336, at least one plugin database 338, at least one collector automator

340, at least one machine manager 342, at least one collector database manager 344, at least one account manager 346, and at least one machine collection manager 348. The collector core 332 communicates with and/or manages each of the collector metric receiver 334, the collector synchronizer 336, the plugin database 338, the collector automator 340, the machine manager 342, the collector database manager 344, the account manager 346, and the machine collection manager 348.

The collector metric receiver 334 is configured to receive metrics/attributes representative of the usage of the computing resources 108 of one or more users and/or workloads. In some cases, the collector metric receiver 334 may include one or more of the calculators 305.

At least a portion of the resource usage may be associated with one or more users. For example, at least a portion of the resource usage may be associated with one or more accounts 350 using the account manager 346. The account 350 may be associated with one or more resource consuming users 352. As a result, the resource usage of multiple accounts 350 and/or resource consuming users 352 may be simultaneously monitored. For example, a corporate entity may have an account 350 with which its employees (e.g., resource consuming users 352) are associated. As a result, as the employees access the computing resources 108, the usage is associated with the account 350 that corresponds to the corporate entity. By associating the usage with the accounts 350 a single metric/attribute collector 303 may be used to monitor multiple accounts 350, which may reduce resources consumed when compared to associating a metric/attribute collector 303 with each account 350 and/or resource consuming user 352. In some cases, each account 350 may be associated with a respective analysis platform 312 and/or a respective metric/attribute collector 303.

The account 350 may be associated with one or more machines 354 (e.g., virtual or physical) that are accessible to the resource consuming users 352 and operate using the computing resources 108. The machines 354 may include, for example, one or more of the operating system(s) 112 and/or the application(s) 114. For example, a machine 354 may be configured to execute a LINUX based operating system, a WINDOWS based operating system, a MAC based operating system, and/or any other operating system in a virtualized or physical environment. In some cases, a machine 354 may be configured to execute one or more workloads 110 (e.g., application(s) 114) without the execution of a guest operating system. For example, a machine 354 may be configured to execute one or more applications 114 directly on the physical machine without any form of virtualization or execute a container and the one or more applications 114 may be configured to execute within the container. Therefore, a container may generally be described as an isolated environment configured to execute on one or more kernels of the machine 354. A container may be executed on a physical or virtual machine.

The machine collection manager 348 allows for a plurality of machines 354 to be associated with at least one machine collection 356. The machine collection manager 348 may also allow for a plurality of containers or uniquely identified (e.g., tagged) workloads to be associated with at least one collection. In some cases, multiple machine collections 356 may be associated with a respective account 350. For example, an owner (or authorized user) of a respective account 350 may add one or more machines 354 to a respective machine collection 356. Each machine collection 356 may include a plurality of machines 354, each configured to execute, for example, a different operating

system 112 and/or application 114 (e.g., the application 114 may be executed using a container).

The plugin database 338 can be configured to be accessed by the metric/attribute collector 303 and may include one or more plugins 358 (e.g., the plugin 358 may be a driver, a listener, a publisher, or the like) that allow the metric/attribute collector 303 to communicate with one or more computing systems 104 such that resource usage can be, for example, measured. Each computing system 104 may be configured to include one or more machines 354 that are each configured to execute different operating systems 112 and/or applications 114. In other words, the plugins 358 allow the metric/attribute collector 303 to monitor resource usage across multiple different computing environments simultaneously. For example, the plugins 358 may include a LINUX plugin, a WINDOWS plugin, a MAC plugin, a SOLARIS plugin, a VMWARE plugin, a HYPER-V plugin, a CITRIX XEN plugin, an OPENSTACK plugin, an OPENSHIFT plugin, a KUBERNETES plugin, a DOCKER plugin, a PIVOTAL plugin, an AMAZON plugin (e.g., an AMAZON WEB SERVICES plugin), a MICROSOFT plugin (e.g., a MICROSOFT AZURE plugin), a GOOGLE plugin (e.g., a GOOGLE CLOUD PLATFORM plugin), a bare metal plugin (e.g., a plugin configured to communicate directly with a non-virtualized machine), an intelligent platform management interface (IPMI) plugin, a simple network management protocol (SNMP) plugin, a basic input/output (BIOS) plugin, a baseboard management controller (BMC) plugin, and/or any other plugin that allows for the monitoring of resource usage across multiple operating systems/environments. The plugins 358 may also allow for the monitoring of one or more virtualized and/or physical environments simultaneously. The plugins 358 may also enable monitoring of other user account information including billing information (e.g., cost and/or pricing information), software program utilization, software licenses rented, computing system information (e.g., computing system make(s), model(s), age, serial number(s), power consumption, fan speed, temperature, configuration quantities and details such as, for example, memory type, processor type, graphics processing system type, and power supply type, and/or any other computing system information) and/or any other additional user account information. In some cases, a plurality of plugins 358 may be simultaneously monitoring resource usage and/or other user information.

The plugins 358 can be configured to be activated (e.g., the metric/attribute collector 303 is able to communicate with a respective computing system 104 using a respective plugin 358) or deactivated (e.g., the metric/attribute collector 303 is unable to communicate with a respective computing system 104 using a respective plugin 358).

In some cases, the metric/attribute collector 303 and/or the analysis platform 312 may utilize one or more plugins to monitor usage and track associated resource costs. The metric/attribute collector 303 may communicate with a computing system via one or more plugins to monitor usage, and the analysis platform 312 may communicate with a resource supplier's API via a one or more different plugins to gather associated costs and related billing details (e.g., server instance types used, time consumed, server pricing, services used, and other related cost information). Furthermore, the metric/attribute collector 303 may track intervals of usage time (e.g., in seconds, minutes, hours, or the like). The metric/attribute collector 303 may also capture server metadata (e.g., server instance names, server instance types, and other server metadata). In some cases, the usage analyzer 386 may analyze unique server instance types and the

quantity of time that a supplier charged for those unique server types and compare with the server instance metadata and associated intervals of time captured by the metric/attribute collector 303 in order to identify potential discrepancies in billing. In some cases, resource suppliers may inaccurately measure and bill their customers for the wrong services or for the incorrect amount of time or usage of those services. By running the meter 118, a user may be able to validate or invalidate that their resource supplier usage costs are accurate/inaccurate for their level of usage.

The machine manager 342 may allow for the assigning and/or associating of machines 354 to a respective machine collection 356. The machine manager 342 may analyze the machines 354 in the machine collection(s) 356 to determine, for example, the specifications and/or the operating system of the machines 354. The specifications may include, for example, information related to the computing resources 108 (e.g., available memory, storage, and/or the like).

The machine manager 342 may also search for new machines that are to be monitored by the metric/attribute collector 303. The machine manager 342 may detect new machines by searching for new IP addresses that have been associated with a specific account 350 and/or resource consuming user 352.

The collector database manager 344 manages the data received and/or generated by the metric/attribute collector 303. For example, the collector database manager 344 may receive and/or cause to be stored one or more records (e.g., data) relating to the machine(s) 354, the machine collection(s) 356, the account(s) 350, and the resource usage associated therewith (e.g., metrics/attributes). The collector database manager 344 may also delete data. For example, the collector database manager 344 may delete records associated with a respective machine 354, a respective machine collection 356, and/or a respective account 350. In some cases, the collector database manager 344 may truncate records associated with a respective machine 354, a respective machine collection 356, and/or a respective account 350. Deletion and/or truncation may occur after, for example, a period of time has elapsed and/or after the deactivation/termination of a respective account 350.

The collector database manager 344 may communicate with a collector database 364 via a communication link 366 and cause one or more records to be stored on the collector database 364. As shown, the collector database 364 may be remote from the metric/attribute collector 303 (e.g., operating on a different server and/or a different network). However, in some cases, the collector database 364 may be local to the metric/attribute collector 303 (e.g., operating on the same server, on the same network, and/or on the same device). Data transmitted via the communication link 366 may be encrypted to prevent unauthorized access to the data during transmission. In some cases, the records stored on the collector database 364 may be encrypted to prevent unauthorized access to the records. For example, the records stored on the collector database 364 may be encrypted using transparent data encryption (TDE). The collector database 364 may be configured to store one or more of metrics/attributes collected by the metric/attribute collector and/or may receive metrics/attributes in the form of a data file such as a JavaScript Object Notation (JSON) file, a Comma Separated Values (CSV) file, a spreadsheet file, and/or any other file type. In some cases, the received data file may be generated independently from the meter 118 by, for example, a user, a third-party application (e.g., via a third-party application programming interface), and/or any other source.

The collector synchronizer 336 may communicate with, for example, the collector database manager 344 and may copy and/or backup data received by the collector database manager 344. For example, the collector synchronizer 336 may copy one or more records stored on and/or to be stored on the collector database 364 to a collector backup 368. In some cases, the records stored on the collector backup 368 may be encrypted to prevent unauthorized access. When a change is made to one or more records, the collector synchronizer 336 may copy and/or backup only the changes to the collector backup 368. As a result, the collector synchronizer 336 may not copy an entire record to the collector backup 368. As shown, the collector synchronizer 336 may communicate with the collector backup 368 via a communication link 370. Data transmitted via the communication link 370 may be encrypted to prevent unauthorized access to the data during transmission. As shown, the collector backup 368 may be remote from the metric/attribute collector 303 (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the collector backup 368 may be local to the metric/attribute collector 303 (e.g., operating on the same server, on the same network, and/or on the same device).

In some cases, at least one backup collector 397 may be provided. The backup collector 397 may operate substantially the same as the metric/attribute collector 303 and may be configured to run in parallel with (or simultaneously with) the metric/attribute collector 303. In some cases, the backup collector 397 is configured to be remote from the metric/attribute collector 303 (e.g., operating on a different server, in a different network, and/or on a different device). As a result, in the event of a failure of the metric/attribute collector 303, the meter 118 may continue to monitor resource usage. In other words, the backup collector 397 may generally be described as providing redundancy. In some cases, there may be two or more backup collectors 397 such that, for example, n+2 redundancy can be achieved. Therefore, the backup collector 397 can include any one or more of the features discussed herein in relation to the metric/attribute collector 303. The backup collector 397 may communicate with the computing system 104 via a communication link 399. Data transmitted via the communication link 399 may be encrypted to prevent unauthorized access to the data during transmission.

The backup collector 397 may also be configured to generate independent records. The independent records may be used during, for example, an auditing process and/or in a data recovery process (e.g., in the event that the metric/attribute collector 303 experiences a failure). In some cases, the collector backup 368 may include the backup collector 397.

As shown, the metric/attribute collector 303 may also include the collector automator 340. The collector automator 340 may automate one or more features of the metric/attribute collector 303. For example, the collector automator 340 may be configured to receive a triggering event that causes metric/attribute collector 303 to carry out a predetermined action. For example, the predetermined action may include one or more of the enabling or disabling access to the computing resources 108, the activating or deactivating of at least a portion of the metric/attribute collector 303 (e.g., the metric receiver 334) or at least a portion of another metric/attribute collector, activating or deactivating one or more plugins 358 (e.g., one or more plugins 358 that are configured to enable communication with the computing system 104), commencing a database cleanup, activating or

deactivating one or more application programming interfaces (API), activating or deactivating at least a portion of the analysis platform 312 or at least a portion of another analysis platform, activating or deactivating at least a portion of another metric/attribute collector 303, commencing or terminating receiving metrics/attributes, and/or any other action. A triggering event may include, for example, discovering new machine(s), expiration of a contractual term, using a predetermined quantity of resources, adding a new machine to a respective machine collection 356, and/or the like. In some cases, the trigger event may be the expiration of a period of time. For example, each one, two, three, four, or five seconds (or any other suitable time interval), the metric/attribute collector 303 may search for new Internet Protocol (IP) addresses (and/or any other identifiers) that correspond to a new machine. In these cases, the metric/attribute collector 303 may generally be described as self-updating.

As also shown, an administrator user interface 372 may be communicatively coupled to the collector core 332. The administrator user interface 372, may provide an interface for an operator of the metric/attribute collector 303 to control and/or modify the metric/attribute collector 303 and/or review data generated and/or received by the metric/attribute collector 303. For example, the administrator user interface 372 may be configured to display and/or cause to be generated usage summaries on a per account 350 or resource consuming user 352 basis. The administrator user interface 372 may also be used to, for example, manually add one or more machines 354, adjust and/or delete one or more records, add or remove new accounts 350, add or remove plugins 358, adjust/create triggers for the collector automator 340, continue and/or discontinue monitoring resource usage at predetermined times in response to, for example, the terms of a contract, and/or any other administrative task.

As shown, the administrator user interface 372 may communicate with the collector core 332 via a communication link 374. Data transmitted via the communication link 374 may be encrypted to prevent unauthorized access to the data during transmission. As shown, the administrator user interface 372 may be remote from the metric/attribute collector 303 (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the administrator user interface 372 may be local to the metric/attribute collector 303 (e.g., operating on the same server, on the same network, and/or on the same device).

In some cases, for example, the administrative user interface 372 may be generated and displayed on a web browser. By way of further example, in some cases, the administrative user interface 372 may be an application executing on a computer that communicates with the metric/attribute collector 303 through a network connection (e.g., the internet). By way of still further example, in some cases, the administrative user interface 372 may be an application executing locally (e.g., on the same device as the metric/attribute collector 303).

FIG. 3C shows an example of an analysis platform 360, which may be an example of the analysis platform 312 of FIG. 3A. The analysis platform 360 may generally be described as being configured to normalize (e.g., convert) a plurality of metrics/attributes that are received over a period of time to a normalized unit and sum the normalized metrics/attributes to generate a total resource usage value represented as a single normalized unit for the period of time. In other words, the analysis platform 360 may be

configured to generate a single usage value that is representative of the total resource usage over a period of time. The analysis platform 360 may include any combination of hardware, software, and/or firmware. In some cases, the analysis platform 360 may be executed on the same server and/or on the same network as the metric/attribute collector 302. In other cases, the analysis platform 360 may be executed on a different server and/or a different network from the metric/attribute collector 302 (e.g., on a user's network and/or server).

The analysis platform 360 may be configured to receive one or more metrics/attributes representative of the usage of the computing resources 108 from one or more of the metric/attribute collector 302 and/or one or more backup collectors 397. In some cases, the analysis platform 360 may receive one or more metrics/attributes from the computing system 104. In these cases, the meter 118 may not include the metric/attribute collector 302 and the analysis platform 360 may include the plugin database 338. The analysis platform 360 may receive the metrics/attributes in the form of, for example, a JavaScript Object Notation (JSON) file, a Comma Separated Values (CSV) file, and/or any other file type.

The analysis platform 360 may be configured to be included with any one or more of the computing system 104, the device 102, and/or any other device communicatively coupled to the network 106. For example, the device 102 may include the analysis platform 360. When the device 102 includes the analysis platform 360, a user of the analysis platform 360 may be able to utilize the meter 118 with fewer modifications to the user's network security, when compared to when the analysis platform 360 is not executed on the device 102. In some cases, when the device 102 includes the analysis platform 360, the analysis platform 360 can be configured to analyze the resource usage of one or more users on, for example, a local network, each of the users having, for example, a respective device. Additionally, or alternatively, when the device 102 includes both the analysis platform 360 and the metric/attribute collector 302, only metrics/attributes representative of the usage of the computing resources 108 may need to be transmitted over public networks (e.g., the internet).

The analysis platform 360 may include at least one analysis core 376, at least one analysis metric receiver 378, at least one analysis synchronizer 380, at least one analysis database manager 382, at least one analysis automator 384, at least one usage analyzer 386, at least one notification generator 388, at least one subscription manager 390, and at least one exchange/contract manager 393. The analysis core 376 may generally be described as communicating with and/or managing each of the analysis metric receiver 378, the analysis synchronizer 380, the analysis database manager 382, the analysis automator 384, the usage analyzer 386, the notification generator 388, and the subscription manager 390.

As shown, the analysis core 376 is communicatively coupled to the analysis metric receiver 378. The analysis metric receiver 378 receives the metrics/attributes representative of the usage of the computing resources 108 from the metric/attribute collector 302. The analysis metric receiver 378 may include one or more of the calculators 305. In some cases, the analysis metric receiver 378 may be configured to generate a total resource usage value represented as a single normalized unit, which may be received by the analysis core 376.

The analysis core 376 and/or the analysis metric receiver 378 may be in communication with the usage analyzer 386.

The usage analyzer 386 monitors usage associated with a respective account 350 and/or resource consuming user 352 to develop (or determine) a normal usage, a peak usage, and/or a trough usage. In other words, the normal usage, peak usage, and/or trough usage may be based, at least in part, on received metrics/attributes corresponding to respective ones of the computing resources 108. In some cases, for example, the peak usage may represent a maximum usage over a period of time (e.g., an hour, a day, a week, a month, and/or any other suitable period of time), the trough usage may represent the minimum usage over a period of time (e.g., an hour, a day, a week, a month, and/or any other suitable period of time), and the normal usage may represent an average usage over a period of time (e.g., an hour, a day, a week, a month, and/or any other suitable period of time). In some cases, the peak usage may approximate the maximum usage over a period of time (e.g., be a usage value within 5%, 10%, or 15% of the actual maximum usage) and trough usage may approximate the minimum usage over a period of time (e.g., be a usage value within 5%, 10%, or 15% of the actual minimum usage). In some cases, the usage analyzer 386 may be incorporated into the metric/attribute collector 302.

The normal usage, the peak usage, and/or the trough usage may correspond to individual computing resources, to a plurality of computing resources, and/or one or more sub-resources (e.g., one or more of a graphics processor and graphics memory of the GPU 210) making up a computing resource. For example, a normal usage, a peak usage, and/or a trough usage may be generated for each of a plurality of computing resources (e.g., each of the CPU 202, memory 204, storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209). In these cases, the normal usage, the peak usage, and/or the trough usage for each resource may be represented as a normalized unit and/or as non-normalized unit(s) (e.g., as units that correspond to the individual resources). By way of further example, a normal usage, a peak usage, and/or a trough usage may be generated that corresponds to a plurality of computing resources (e.g., two or more of the CPU 202, memory 204, storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209). In these cases, the normal usage, the peak usage, and/or the trough usage may be represented as a single normalized unit.

In some cases, the usage analyzer 386 may be configured to generate an overall peak usage average. The overall peak usage average may generally be described as the average of two or more peak usage values corresponding to two or more respective predetermined periods of time. For example, the peak usage may be determined for each day of a week and averaged to determine an overall peak usage average for a week. Additionally, or alternatively, the usage analyzer 386 may be configured to generate an overall trough usage average. The overall trough usage average may generally be described as the average of two or more trough usage values corresponding to two or more respective predetermined periods of time. For example, the trough usage may be determined for each day of a week and averaged to determine an overall trough usage average for a week.

The overall peak usage average and/or the overall trough usage average may correspond to individual computing resources and/or to a plurality of computing resources. For example, an overall peak usage average and/or an overall trough usage average may be generated for each of a plurality of computing resources (e.g., each of the CPU 202, the memory 204, the storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources

209). In these cases, the overall peak usage average and/or the overall trough usage average for each resource may be represented as a normalized unit and/or as non-normalized unit(s) (e.g., as units that correspond to the individual resources). By way of further example, an overall peak usage average and/or an overall trough usage average may be generated that corresponds to a plurality of computing resources (e.g., two or more of the CPU 202, the memory 204, the storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209). In these cases, the overall peak usage average and/or the overall trough usage average may be represented as a single normalized unit.

A normal average usage may generally be described as an average resource usage that is calculated by taking multiple measurements of an actual resource usage over a period of time within repeating time intervals and averaging the measurements. For example, the actual resource usage may be measured at 11:00 AM every day for five days and the average usage for 11:00 AM may be calculated based on these measurements. A peak average usage may generally be described as an average resource usage that is calculated by taking multiple measurements of an actual resource usage over a period of time within repeating time intervals, selecting a portion of the measurements having the highest usages (e.g., the highest 5%, 10%, 15%, 20%, 25%, 35%, 40%, 50%, or 60% of the resource usage), and calculating the average of the selected portion. For example, the actual resource usage may be measured at 11:00 AM every day for five days and the peak average usage for 11:00 AM may be calculated based on the three highest measurements. A trough average usage may generally be described as an average resource usage that is calculated by taking multiple measurements of an actual resource usage over a period of time within repeating time intervals, selecting a portion of the measurements having the lowest usages (e.g., the lowest 5%, 10%, 15%, 20%, 25%, 35%, 40%, 50%, or 60% of the resource usage), and calculating the average of the selected portion. For example, the actual resource usage may be measured at 11:00 AM every day for five days and the trough average usage for 11:00 AM may be calculated based on the three lowest measurements.

The normal average usage, the peak average usage, and/or the trough average usage may correspond to individual computing resources and/or to a plurality of computing resources. For example, a normal average usage, a peak average usage, and/or a trough average usage may be generated for each of a plurality of computing resources (e.g., each of the CPU 202, the memory 204, the storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209). In these cases, the normal average usage, the peak average usage, and/or the trough average usage for each resource may be represented as a normalized unit and/or as non-normalized unit(s) (e.g., as units that correspond to the individual resources). By way of further example, a normal average usage, a peak average usage, and/or a trough average usage may be generated that corresponds to a plurality of computing resources (e.g., two or more of the CPU 202, the memory 204, the storage 206, the network interface 208, the GPU 210, and/or one or more of the other resources 209). In these cases, the normal average usage, the peak average usage, and/or the trough average usage may be represented as a single normalized unit.

The usage analyzer 386 may generate a normal average usage pattern 392, a peak average usage pattern 394, and a trough average usage pattern 395 (illustrated graphically in

FIG. 3D for purposes of clarity) based on the normal average usage, the peak average usage, and the trough average usage, respectively. The normal average usage pattern **392** may be a data set comprising a plurality of normal average usage values that correspond to different predetermined times (or predetermined periods of time). The peak average usage pattern **394** may be a data set comprising a plurality of peak average usage values that correspond to different predetermined times (or predetermined periods of time). The trough average usage pattern **395** may be a data set comprising a plurality of trough average usage values that correspond to different predetermined times (or predetermined periods of time).

The data set corresponding to the normal average usage pattern **392** may be compared to the data set corresponding to the peak average usage pattern **394** to determine a threshold usage value or range for one or more predetermined times (or predetermined periods of time). For example, a threshold usage value for a predetermined time may be obtained by taking the average of the normal average usage value and the peak average usage value at a respective predetermined time. By way of further example, a threshold usage range may be described as the range extending from the normal average usage value to the peak average usage value at a respective predetermined time.

Additionally, or alternatively, the data set corresponding to the trough average usage pattern **395** may be compared to the data set corresponding to the peak average usage pattern **394** to determine a threshold usage value or range for one or more predetermined times (or predetermined periods of time). For example, a threshold usage value for a predetermined time may be obtained by taking the average of the trough average usage value and the peak average usage value at a respective predetermined time. By way of further example, a threshold usage range may be described as the range extending from the trough average usage value to the peak average usage value at a respective predetermined time.

Additionally, or alternatively, the data set corresponding to the normal average usage pattern **392** may be compared to the data set corresponding to the trough average usage pattern **395** to determine a threshold usage value or range for one or more predetermined times (or predetermined periods of time). For example, a threshold usage value for a predetermined time may be obtained by taking the average of the normal average usage value and the trough average usage value at a respective predetermined time. By way of further example, a threshold usage range may be described as the range extending from the trough average usage value to the normal average usage value at a respective predetermined time.

Additionally, or alternatively, the threshold usage value may be a trough average usage value, a peak average usage value, a trough usage value (i.e., a minimum usage over a predetermined time) or an overall trough usage average value, and/or a peak usage value (i.e., a maximum usage over a predetermined time) or an overall peak usage average value.

The notification generator **388** may be configured to cause a notification to be generated in response to, for example, one or more of resource usage at a predetermined time (or predetermined time interval) meeting, exceeding, and/or failing to exceed a threshold usage value and/or resource usage falling outside of a threshold usage range. In other words, a notification may be caused to be generated based, at least in part, on a comparison of the resource usage to a threshold or a threshold usage range. For example, detecting

a usage that exceeds the threshold usage value may be indicative of an unauthorized third-party using the resources and detecting a usage that does not exceed the trough average usage value may be indicative of a hardware failure (e.g., at least a portion of the computing resources **108** may be malfunctioning).

In some cases, the usage analyzer **386** may be configured to identify outliers (e.g., resource usage that falls outside of the threshold usage range). For example, the usage analyzer **386** may utilize machine learning to identify periods of time of high and low usage. In periods of high usage and/or low usage, the usage analyzer **386** may identify usage as an outlier if the usage does not, for at least a period of time (e.g., 10 minutes, 30 minutes, 60 minutes, 2 hours, and/or any other time span), exceed the peak average usage value and/or fails to exceed the trough average usage value. As a result, the notification generator **388** may be configured to not generate a notification in response to the detection of an outlier. Further, when an outlier is identified, the usage analyzer **386** may exclude the outlier from the calculation of the normal average usage pattern **392**, the peak average usage pattern **394**, and/or the trough average usage pattern **395** unless, for example, the outlier is detected multiple times. Therefore, the normal average usage pattern **392**, the peak average usage pattern **394**, and/or the trough average usage pattern **395** may each be generally described as being generated using machine learning.

In some cases, the usage analyzer **386** may analyze the usage of other users of the meter **118** to determine if other users are experiencing similar usage patterns. For example, if multiple users are experiencing a resource usage that does not exceed the trough average resource value, it may be indicative of a hardware failure (e.g., at least a portion of the computing resources **108** may be malfunctioning). In this case, the notification generator **388** may cause a notification to be generated.

The generated notification may be in the form of an alert transmitted to an owner/user of, for example, one or more of the accounts **350** via, for example, telephone call, SMS text message, e-mail, fax, tactile sensation (e.g., via a smart watch, fitness tracker, remote health monitoring device, and/or any other wearable device), an alert generated within an application or software for managing the analysis platform **360** and/or the metric/attribute collector **302** (e.g., an in-application message), and/or any other type of alert.

The usage analyzer **386** may determine periods of high resource usage and periods of low resource usage that occur on a periodic basis over a period of time using, for example, machine learning. When the periods of high resource usage and low resource usage are identified, the threshold usage at which, for example, an alert is caused to be generated may be adjusted. For example, during a known period of high resource usage the threshold usage for generation of an alert may be adjusted to be higher than during a known period of low resource usage.

The usage analyzer **386** may also be configured to generate a benchmark (e.g., an average cost index) representative of a per unit cost of consumed resources for one or more users and/or accounts. The usage analyzer **386** may receive a total resource usage value represented as a single normalized unit associated with one or more accounts **350**, resource consuming users **352**, and/or machine collections **356**. In some cases, the usage analyzer **386** may receive a total resource usage value that corresponds to a workload type (e.g., SPARK cluster workloads, ELASTICSEARCH workloads, data warehousing workloads, disaster recovery workloads, developer workloads, quality assurance workloads,

production workloads, and/or the like), a service provider (e.g., AMAZON, MICROSOFT, GOOGLE, and/or any other service provider), a hardware provider (e.g., DELL, SUPERMICRO, HP, and/or any other hardware provider), a hardware configuration, a hypervisor (e.g., VMWARE, CITRIX, OPENSTACK, and/or any other hypervisor), and/or a group (e.g., a business unit within an organization such as an engineering, sales, or marketing department).

The usage analyzer 386 may also receive cost data associated with the resources being consumed. The cost data may be received from the computing system 104 (e.g., the service provider), from user input, and/or any other source. The cost data may be representative of the cost of the total resource usage associated with a respective account 350, resource consuming user 352, machine collection 356, workload type, and/or group. The cost data associated with the resources being consumed may include one of more of a plurality of colocation costs, hardware costs, networking costs, cloud hosting costs, labor costs, software license costs, compliance costs, managed services costs, consultant costs, support costs, developer/operations education costs, cybersecurity costs, migration opportunity costs, and/or any other associated cost for resources being consumed. The usage analyzer 386 may then divide the cost of the total resource usage by the total resource usage value to obtain a cost per unit consumed value. The usage analyzer 386 may utilize the cost per unit consumed to generate a benchmark (e.g., an average cost index).

The cost per unit consumed value may then be compared to, for example, a market benchmark (e.g., an average per unit cost across all users of one or more meters 118), a group or industry benchmark (e.g., an average per unit cost across all users within a predetermined industry or group), a workload benchmark (e.g., an average per unit cost across all users executing predetermined workloads), a service provider benchmark (e.g., AMAZON, MICROSOFT, GOOGLE, and/or any other service provider), and/or the like. The group or industry benchmark may be based on, for example, company size (e.g., market cap, annual revenues, and/or the like), business units or departments within a specific company or across multiple companies (e.g., a sales department, an engineering department, and/or the like), and/or any other groupings.

In some cases, the usage analyzer 386 may also be configured to generate the market benchmark, the group or industry benchmark, the service provider benchmark, and/or the workload benchmark. For example, the usage analyzer 386 may be configured to access data relating to resource usage and resource costs of one or more accounts 350 that correspond to one or more different users.

In some cases, the usage analyzer 386 may be separate from the analysis platform 360. For example, the usage analyzer 386 may be configured to communicate with multiple analysis platforms 360 and/or metric/attribute collectors 302. In some cases, there may be a market usage analyzer that is remote from the analysis platform 360 and that is configured to determine average usage and average cost for specific groups or industries, specific accounts, specific workloads, and the like across multiple different users. In these cases, there may also be a local usage analyzer that is configured to determine an average usage and an average cost for a specific analysis platform 360.

In some cases, the usage analyzer 386 can also be configured to analyze the computing resources 108 being consumed to determine whether the workload 110 being executed on the computing resources 108 is utilizing the computing resources 108 efficiently. For example, the usage

analyzer 386 may analyze the specifications of the computing resources 108 (e.g., processor speed and/or size, memory speed and/or size, GPU speed and/or size, storage speed and/or size, and/or the like) and compare them to the requirements of the workloads 110 such that the usage analyzer 386 can generate an optimized computing resources configuration (e.g., a best estimated configuration). In this case, the usage analyzer 386 can be configured to cause the generation of a notification to be sent to a user, wherein the notification can include a description of the optimized computing resources configuration. In some cases, the usage analyzer 386 can be configured to automatically transition the workloads 110 to the optimized computing resources configuration. By way of further example, the computing resources 108 may include a dedicated graphics processing unit, however, the applications 114 and/or the operating systems 112 executing on the computing resources 108 may not need a dedicated graphics processing unit to operate efficiently. In this case, the usage analyzer 386 can be configured to cause the generation of a notification to be sent to a user. In some cases, the usage analyzer 386 can be configured to automatically move the workload 110 to another computing system 104 that does not include a dedicated graphics processing unit. As a result, the cost per unit of computing resources consumed may be decreased.

In some cases, the usage analyzer 386 may further be configured to generate computing resource configuration recommendations based on the workloads 110 to be executed. The computing resource configuration recommendations may be associated with a cost per unit of computing resources consumed value and/or an estimated performance rating for a respective workload 110. As a result, a user may compare resource costs and/or performance between multiple resource suppliers and/or multiple computer resources options available from a particular resource supplier.

In some cases, the usage analyzer 386 may be configured to analyze a user's resource usage when the user is purchasing a predetermined quantity of resources in advance. In these cases, the usage analyzer 386 may convert the quantity of pre-purchased resources to a normalized value representative of a maximum quantity of resources available and compare the normalized value of the pre-purchased resources to the peak usage value. In other words, the usage analyzer 386 can be configured to determine if a resource supply is sufficient to meet a user's peak resource demand when the user pre-purchases a predetermined quantity of resources. Therefore, based on this comparison, the usage analyzer 386 may be configured to provide recommendations regarding a quantity of resources to pre-purchase.

The subscription manager 390 manages and maintains subscription information relating to users of the meter 118. For example, the subscriptions may determine how many resource consuming users 352 may be assigned to a respective account 350, determine how many accounts may be assigned to a respective user (e.g., a corporate entity), determine how many machines 354 that may be accessed by a respective user, determine how many machines 354 that may be assigned to a respective machine collection 356, determine how many machine collections 356 a respective user may have, associate login information with respective users, determine validity of login information (e.g., whether or not the login information has expired), and/or any other task associated with subscription management.

The exchange/contract manager 393 is configured to reconcile contracts between two parties (e.g., a buyer and a seller). The exchange/contract manager 393 may be config-

ured to deploy, disable, start, and/or stop the meter **118** in accordance with the terms of a contract. When the meter **118** has been disabled, a user's access to the computing resources **108** may be disabled. The exchange/contract manager **393** may also be configured to audit the resource usage. The auditing of the resource usage may involve confirming that the resources indicated as being used were actually used and/or that the usage was in accordance with the contract. The auditing process may involve comparing data gathered by the metric/attribute collector **302** to data generated by a second collector (e.g., a backup collector). The auditing process may also include confirming that a resource supplier is providing an appropriate quantity and/or quality of computing systems and/or resources to a buyer (e.g., by monitoring a quantity of memory or storage, type of processor, specified hardware manufacturer, and/or any other information agreed upon between buyer and seller).

The analysis automator **384** may automate one or more features of the analysis platform **360**. For example, the analysis automator **384** may be configured to receive a triggering event that causes the analysis platform **360** to carry out a predetermined action. The predetermined action may include, for example, one or more of the enabling or disabling access to the computing resources **108**, the activating or deactivating of at least a portion of the metric/attribute collector **302** (e.g., the metric receiver **334**) or at least a portion of another metric/attribute collector, activating or deactivating one or more plugins **358** (e.g., one or more plugins **358** that are configured to enable communication with the computing system **104**), commencing a database cleanup, activating or deactivating one or more application programming interfaces (API), activating or deactivating at least a portion of the analysis platform **360** or at least a portion of another analysis platform **360**, activating or deactivating at least a portion of another analysis platform **360**, and/or any other action. A triggering event may include, for example, receiving a new contract, receiving a new subscription, a user input from the administrator user interface **372**, a predetermined time (e.g., the start of a new contract/subscription and/or any other predetermined time), a recall request, and/or the like. In some cases, the triggering event may be the expiration of a period of time (e.g., the lapsing of a contractual term). For example, the analysis platform **360** may carry out a predetermined action in response to a contractual term lapsing.

As shown, the analysis database manager **382** communicates with the analysis core **376** and the analysis synchronizer **380**. The analysis database manager **382** manages the data received and/or generated by the analysis platform **360**. For example, the analysis database manager may manage and/or cause to be stored one or more records (e.g., data) relating to subscriptions, contracts, metrics/attributes, usage patterns (e.g., the normal average usage pattern **392**, the peak average usage pattern **394**, or the trough average usage pattern **395**), notification thresholds, triggering events, and/or any other data received or generated by the analysis platform **360**. The analysis database manager **382** may also delete and/or truncate data. For example, the analysis database manager **382** may delete and/or truncate records relating to subscriptions, contracts, metrics/attributes, usage patterns (e.g., the normal average usage pattern **392**, the peak average usage pattern **394**, or the trough average usage pattern **395**), notification thresholds, and/or any other data received by the analysis platform **360**. Deletion and/or truncation may occur after, for example, a period of time has elapsed and/or after the deactivation/termination of, for example, a subscription or contract.

The analysis database manager **382** may communicate with an analysis database **396** via a communication link **398** and cause one or more records to be stored on the analysis database **396**. As shown, the analysis database **396** may be remote from the analysis platform **360** (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the analysis database **396** may be local to the analysis platform **360** (e.g., operating on the same server, on the same network, and/or on the same device). Data transmitted via the communication link **398** may be encrypted to prevent unauthorized access during transmission. In some cases, the records stored on the analysis database **396** may be encrypted to prevent unauthorized access to the records. For example, the records stored on the analysis database **396** may be encrypted using transparent data encryption (TDE). The analysis database **396** may be configured to store one or more of metrics/attributes collected by the metric/attribute collector and/or may receive metrics/attributes in the form of a data file such as a JavaScript Object Notation (JSON) file, a Comma Separated Values (CSV) file, a spreadsheet file, and/or any other file type. In some cases, the received data file may be generated independently from the meter **118** by, for example, a user, a third-party application (e.g., via a third-party application programming interface), and/or any other source.

The analysis synchronizer **380** may communicate with the analysis metric receiver **378** and/or the analysis database manager **382** and may copy and/or backup data received by the analysis platform **360** (e.g., the data received by the analysis database manager **382**). For example, the analysis synchronizer **380** may copy and/or backup one or more of the records stored on and/or to be stored on the analysis database **396** to an analysis backup **381**. In some cases, the records stored on the analysis backup **381** may be encrypted to prevent unauthorized access. When a change is made to one or more of the records, the analysis synchronizer **380** may backup only the changes to the analysis backup **381**. As shown, the analysis synchronizer **380** may communicate with the analysis backup **381** via a communication link **383**. Data transmitted via the communication link **383** may be encrypted to prevent unauthorized access to the data during transmission. As shown, the analysis backup **381** may be remote from the analysis platform **360** (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the analysis backup **381** may be local to the analysis platform **360** (e.g., operating on the same server, on the same network, and/or on the same device).

In some cases, a backup analysis platform **355** that operates substantially the same as the analysis platform **360** may be provided. As a result, independent records may be maintained. The independent records may be used, for example, in an auditing process and/or in a data recovery process (e.g., in the event that the analysis platform **360** experiences a failure). The backup analysis platform **355** may be communicatively coupled to the metric/attribute collector **302** using a communication link **353**.

In some cases, the analysis backup **381** may also include a backup analysis platform that operates substantially the same as the analysis platform **360**. In other words, the analysis backup **381** may include a standalone analysis platform. As a result, independent records may be maintained. The independent records may be used, for example, in an auditing process and/or in a data recovery process (e.g., in the event that the analysis platform **360** experiences a failure).

As shown, a usage forecaster **385** may be communicatively coupled to the analysis core **376** via a communication link **387**. The usage forecaster **385** may estimate future usage for a respective user (e.g., that owns or controls one or more accounts **350**), a respective account **350**, and/or a respective resource consuming user **352**. The usage forecaster **385** may use, for example, the normal average usage pattern **392**, the peak average usage pattern **394**, and/or the trough average usage pattern **395** to forecast a future usage. Additionally, or alternatively, the usage forecaster **385** may analyze historical resource usage utilizing machine learning to forecast a future usage. For example, the usage forecaster **385** may analyze the usage for patterns and/or surges (e.g., an increase in resource usage). Once the future usage is forecasted, the usage forecaster **385** may also estimate a forecasted economic cost that corresponds to the forecasted usage.

As shown, an analysis user interface **389** may communicate with the analysis core **376** via a communication link **391**. The analysis user interface **389** may be configured to generate a display that presents information relating to the analysis platform **360** and/or the metric/attribute collector **302**. For example, the analysis user interface **389** may be configured to display a usage forecast, the normal average usage pattern **392** (e.g., as a graph or plot), the peak average usage pattern **394** (e.g., as a graph or plot), the trough average usage pattern **395** (e.g., as a graph or plot), current contract terms, current subscriptions, active and/or inactive notifications/alerts, and/or any other data regarding the analysis platform **360**. In some cases, the analysis user interface **389** may be configured to allow a user to modify various features of the analysis platform **360**. For example, the analysis user interface **389** may be configured such that a user can change and/or cancel one or more subscriptions, change and/or cancel one or more contracts, change, add, and/or remove notification thresholds and/or analysis automator **384** triggers, and/or change any other feature of the analysis platform **360**.

As shown, the analysis user interface **389** may be remote from the analysis platform **360** (e.g., operating on a different server, in a different network, and/or on a different device). However, in some cases, the analysis user interface **389** may be local to the analysis platform **360** (e.g., operating on the same server, on the same network, and/or on the same device). Data transmitted over the communication link **391** may be encrypted to prevent unauthorized access to the data.

For example, in some cases, the analysis user interface **389** may be generated and displayed on a web browser. By way of further example, in some cases, the analysis user interface **389** may be an application executing on a computer that communicates with the analysis platform **360** through a network connection (e.g., the internet). By way of still further example, in some cases, the analysis user interface **389** may be an application executing locally (e.g., on the same hardware as the analysis platform **360**).

The communication links **330**, **353**, **362**, **366**, **370**, **374**, **383**, **387**, **391**, **398**, and/or **399** may utilize the secure sockets layer (SSL) security protocol when establishing communications. As discussed herein, the communication links **330**, **353**, **362**, **366**, **370**, **374**, **383**, **387**, **391**, **398**, and/or **399** may each carry encrypted data. Therefore, one or more of the communication links **330**, **353**, **362**, **366**, **370**, **374**, **383**, **387**, **391**, **398**, and/or **399** may generally be described as being secure communication links (e.g., the communication links **330**, **353**, **362**, **366**, **370**, **374**, **383**, **387**, **391**, **398**, and/or **399** use at least one of the SSL security protocol and/or the data transmitted is encrypted).

The metric/attribute collector **302** and the analysis platform **360** may be implemented in software, firmware, hardware, and/or combinations thereof. For example, the metric/attribute collector **302** and the analysis platform **360** may be implemented as software configured to be stored on one or more memories (e.g., any type of tangible, non-transitory storage medium, and may include any one or more of a magnetic recording medium (e.g., a hard disk drive), optical disks, semiconductor devices such as read-only memories (ROMs), random access memories (RAMs) such as dynamic and static RAMs, erasable programmable read-only memories (EPROMs), electrically erasable programmable read-only memories (EEPROMs), flash memories, magnetic or optical cards, or any type of storage media for storing electronic instructions) and executed by one or more processors (e.g., processors offered for sale by INTEL, Advanced Micro Devices, IBM, ARM, Oracle, and/or any other processor) to carry out one or more operations. By way of further example, the metric/attribute collector **302** and the analysis platform **360** may be implemented as circuitry (e.g., an application-specific integrated circuit). The metric/attribute collector **302** and the analysis platform **360** may be implemented on the same or different machine, server, and/or network.

FIG. 4 shows a schematic example of a conversion unit generator **400** configured to generate the conversion units as discussed in relation to FIG. 3A. The conversion unit generator **400** may be separate from the computing system **104**, the device **102**, and the network **106**. For example, the conversion unit generator **400** may be part of a third-party system such that the conversion units may be generated and input into the analysis platform **312** as a fixed value. As shown, the conversion unit generator **400** may include at least a first resource conversion unit generator **402** and a second resource conversion unit generator **404**. The first resource conversion unit generator **402** may generate, for example, one or more of the CPU conversion unit **314**, the memory conversion unit **316**, the storage conversion unit **318**, the network conversion unit **320**, and/or the disk IO conversion unit **328**. The second resource conversion unit generator **404** may generate, for example, one or more of the GPU processor conversion unit **322**, the GPU memory conversion unit **324**, and/or the GPU memory speed conversion unit **326**.

As shown, the first resource conversion unit generator **402** includes a CPU conversion unit generator **406**, a memory conversion unit generator **408**, a storage conversion unit generator **410**, a network conversion unit generator **412**, and a disk IO conversion unit generator **414**.

The CPU conversion unit generator **406** may generate the CPU conversion unit **314** based, at least in part, on a CPU allocation **416** and a CPU compute portion **418**. For example, the CPU conversion unit **314** may be generated by dividing the CPU allocation **416** by the CPU compute portion **418**.

The CPU allocation **416** corresponds to the clock speed of the CPU **202** that is user accessible. For example, only a portion of the physical processor may be allocated to the workload(s) of one or more users. In other words, the usable clock speed allocated may measure less than default clock speed of the physical processor. The CPU compute portion **418** may generally correspond to a weight that is representative of CPU resources used by one or more of the workloads **110**. In some cases, the CPU compute portion **418** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the CPU compute portion **418** may generally be

described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The memory conversion unit generator **408** may generate the memory conversion unit **316** based, at least in part, on a memory allocation **420** and a memory compute portion **422**. For example, the memory conversion unit **316** may be generated by dividing the memory allocation **420** by the memory compute portion **422**.

The memory allocation **420** corresponds to the quantity of the memory **204** that is user accessible. For example, only a portion of the physical memory may be allocated to the workload(s) of one or more users. In other words, the memory allocated may measure less than the total physical memory available. The memory compute portion **422** may generally correspond to a weight that is representative of memory resources used by one or more of the workloads **110**. In some cases, the memory compute portion **422** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the memory compute portion **422** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The storage conversion unit generator **410** may generate the storage conversion unit **318** based, at least in part, on a storage allocation **424** and a storage compute portion **426**. For example, the storage conversion unit **318** may be generated by dividing the storage allocation **424** by the storage compute portion **426**.

The storage allocation **424** corresponds to the quantity of the storage **206** that is user accessible. For example, only a portion of the physical storage may be allocated to the workload(s) of one or more users. In other words, the storage allocated may measure less than the total physical storage available. The storage compute portion **426** may generally correspond to a weight that is representative of storage resources used by one or more of the workloads **110**. In some cases, the storage compute portion **426** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the storage compute portion **426** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The network conversion unit generator **412** may generate the network conversion unit **320** based, at least in part, on a network allocation **428** and a network compute portion **430**. For example, the network conversion unit **320** may be generated by dividing the network allocation **428** by the network compute portion **430**.

The network allocation **428** corresponds to the network bandwidth that is user accessible. For example, only a portion of the total physical network bandwidth may be allocated to the workload(s) of one or more users. In other words, the network bandwidth allocated may measure less than the total physical network bandwidth available. The network compute portion **430** may generally correspond to a weight that is representative of network resources (e.g., bandwidth) used by one or more of the workloads **110**. In some cases, the network compute portion **430** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the network compute portion **430** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The disk IO conversion unit generator **414** may generate the disk IO conversion unit **328** based, at least in part, on a disk IO allocation **432** and a disk IO compute portion **434**.

For example, the disk IO conversion unit **328** may be generated by dividing the disk IO allocation **432** by the disk IO compute portion **434**.

The disk IO allocation **432** corresponds to the disk bandwidth (e.g., reads and writes to the storage **206**) that is user accessible. For example, only a portion of the total physical disk bandwidth may be allocated to the workload(s) of one or more users. In other words, the disk bandwidth allocated may measure less than the total physical disk bandwidth available. The disk IO compute portion **434** may generally correspond to a weight that is representative of disk IO resources (e.g., disk bandwidth) used by one or more of the workloads **110**. In some cases, the disk compute portion **434** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the disk compute portion **434** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The sum of the CPU compute portion **418**, the memory compute portion **422**, the storage compute portion **426**, the network compute portion **430**, and the disk IO compute portion **434** may add up to 100. In other words, each compute portion may be represented as a percentage of the total computing resources evaluated by the first resource conversion unit generator **402** by dividing the compute portion by 100.

As shown, the second resource conversion unit generator **404** includes a GPU processor conversion unit generator **436**, a GPU memory conversion unit generator **438**, and a GPU memory speed conversion unit generator **440**.

The GPU processor conversion unit generator **436** may generate the GPU processor conversion unit **322** based, at least in part, on a GPU processor allocation **442** and a GPU processor compute portion **444**. For example, the GPU processor conversion unit **322** may be generated by dividing the GPU processor allocation **442** by the GPU processor compute portion **444**.

The GPU processor allocation **442** corresponds to the clock speed of the GPU **210** that is user accessible. For example, only a portion of the physical GPU processor may be allocated to the workload(s) of one or more users. In other words, the usable clock speed of the GPU processor that is allocated may measure less than the default clock speed of the physical GPU processor. The GPU processor compute portion **444** may generally correspond to a weight that is representative of GPU processor resources used by one or more of the workloads **110**. In some cases, the GPU compute portion **444** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the GPU compute portion **444** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The GPU memory conversion unit generator **438** may generate the GPU memory conversion unit **324** based, at least in part, on a GPU memory allocation **446** and a GPU memory compute portion **448**. For example, the GPU memory conversion unit **324** may be generated by dividing the GPU memory allocation **446** by the GPU memory compute portion **448**.

The GPU memory allocation **446** corresponds to the quantity of the memory of the GPU **210** that is user accessible. For example, only a portion of the physical memory of the GPU **210** may be allocated to the workload(s) of one or more users. In other words, the memory of the GPU **210** that is allocated may measure less than the total

physical memory of the GPU **210** that is available. The GPU memory compute portion **448** may generally correspond to a weight that is representative of memory resources used by one or more of the workloads **110**. In some cases, the GPU memory compute portion **448** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the GPU memory compute portion **448** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The GPU memory speed conversion unit generator **440** may generate the GPU memory speed conversion unit **326** based, at least in part, on a GPU memory speed allocation **450** and a GPU memory speed compute portion **452**. For example, the GPU memory speed conversion unit **326** may be generated by dividing the GPU memory speed allocation **450** by the GPU memory speed compute portion **452**.

The GPU memory speed allocation **450** corresponds to the clock speed of the GPU memory that is user accessible. For example, only a portion of the clock speed of the physical GPU memory may be allocated to the workload(s) of one or more users. In other words, the usable clock speed of the GPU memory that is allocated may measure less than the default clock speed of the physical GPU memory. The GPU memory speed compute portion **452** may generally correspond to a weight that is representative of GPU memory speed resources used by one or more of the workloads **110**. In some cases, the GPU memory speed compute portion **452** may be an experimentally derived value that is based on an analysis of the workloads of several users. As such, in some cases, the GPU memory speed compute portion **452** may generally be described as being a dynamic weight that is adjustable at predetermined times to account for changes in workloads.

The sum of the GPU processor compute portion **444**, the GPU memory compute portion **448**, and the GPU memory speed compute portion **452** may add up to 100. In other words, each compute portion may be represented as a percentage of the total computing resources evaluated by the second resource conversion unit generator **404** by dividing the compute portion by 100.

As previously discussed, the physical computing resources **108** may additionally, or alternatively, include the other resources **209**. As discussed, the other resources **209** may include any one or more of a Field Programmable Gate Array (FPGA), a Tensor Processing Unit (TPU), an Intelligence Processing Unit (IPU), a Neural Processing Unit (NPU), a Vision Processing Unit (VPU), a Digital Signal Processor (DSP), an Accelerated Processing Unit (APU), an Application-Specific Integrated Circuit (ASIC), a System-on-Chip (SoC), a programmable SoC, an Application Specific Standard Parts (ASSP), an Adaptive Compute Acceleration Platform (ACAP), a microcontroller, and/or any other computing resource. In these cases, the resource conversion units may each be determined using an independent resource conversion unit generator that corresponds to a respective one of the other computing resources **209**. For example, the conversion unit generator **400** can include an other resource conversion unit generator **454**. The other resource conversion unit generator **454** can be configured to generate one or more conversion units that correspond to a respective one or more of the other resources **209**.

FIG. 5 shows an example flow chart of a method **500** for quantifying usage of disparate computing resources as a single unit of measure, which may, for example, be used in the system of FIG. 1. The method **500** may be embodied as one or more instructions in one or more non-transitory

computer readable mediums (e.g., the storage **206**) that are configured to be executed by one or more processors (e.g., the CPU **202**).

As shown, the method **500** may include step **502**. Step **502** may include causing a workload (e.g., the workload **110**) to be executed on a computing system (e.g., the computing system **104**). In some cases, the workload may be caused to be executed, for example, in response to receiving a request from the device **102**. Additionally, or alternatively, the workload may include one or more idle processes caused by operation of the computing system.

The method **500** may also include a step **504**. The step **504** may include measuring a quantity of physical resources used (e.g., consumed) by the execution of the workload over a period of time (e.g., 1 second, 10 seconds, 1 minute, 1 hour, 1 day, 1 week, 1 month, 1 year, and/or any other suitable period of time). The physical quantity of resources may be measured (as opposed to, for example, virtual resources) to prevent or otherwise mitigate inaccurate readings used by the ballooning technique which may be employed by hypervisors measuring virtual resource usage. Further, by measuring the physical resources directly, it may be possible to implement the method on multiple forms of hardware, including hardware not running a virtual machine. Therefore, it is possible to implement this method in systems utilizing packaged software (e.g., containers).

The method **500** may include a step **506**. The step **506** may include normalizing the measurements of the physical resources used. Normalizing the measurements may include converting each of the disparate resources (e.g., CPU resources, memory resources, storage resources, network interface resources, GPU resources, and/or any other computing resource) to a common unit (e.g., as described in relation to FIGS. 3 and 4).

The method **500** may include a step **508**. The step **508** may include summing the normalized measurements. The method **500** may also include a step **510**. The step **510** may include generating a single value representative of the summed normalized resources used. For example, the generated single value may be the summed result. In other cases, the generated value may represent an average of summed normalized resources over a time window (e.g., a weekly average in which the individual summed normalized values correspond to days of the week).

FIG. 6 shows an example flow chart of a method **600** for quantifying a maximum or minimum value of disparate computing resources as a single unit of measure, which may, for example, be used in the system of FIG. 1. The method may be embodied as one or more instructions in one or more non-transitory computer readable mediums (e.g., the storage **206**) that are configured to be executed by one or more processors (e.g., the CPU **202**).

The method may include a step **602**. The step **602** may include measuring, over a period of time, a maximum quantity of physical resources available and/or measuring, over a period of time, a minimum quantity of physical resources for running idle processes (e.g., idle processes **113**).

The method may include a step **604**. The step **604** may include normalizing the measurements of disparate physical resources available (e.g., both used and inactive resources). Normalizing the measurements may include converting each of the disparate resources (e.g., CPU resources, memory resources, storage resources, network interface resources, GPU resources, and/or any other computing resource) to a common unit (e.g., as described in relation to FIGS. 3A-C and 4).

The method may also include a step 606. The step 606 may include summing the normalized measurements of physical resources available. The method may include a step 608. The step 608 may include generating a single value representative of the summed normalized resources available. For example, the generated single value may be the summed result. By way of further example, the generated single value may be an adjusted value that is based on the summed result.

In some cases, a maximum and/or minimum quantity of physical computing resources available over a period of time may be calculated based, at least in part, on the specifications corresponding to individual physical computing resources. This may allow a user to estimate the maximum and/or minimum quantity of physical computing resources available without having access to the physical computing resources. In some cases, the calculated value may be compared to the measured value in the method 600. For example, the calculated value may be compared to the measured value to adjust the specifications that correspond to the individual computing resources and/or the calculation method such that the estimation of the maximum and/or minimum quantity of physical computing resources available may be improved.

FIG. 7 shows an example of a meter portal 700. The meter portal 700 may include and/or provide access to the admin user interface 372 and/or the user interface 389. The meter portal 700 may be implemented in software, firmware, hardware, and/or combinations thereof. For example, the meter portal 700 may be accessible as a webpage to a user of the meter 118. As shown, the meter portal 700 shows an example of resource usage for a plurality of disparate computing resources represented as a single unit of measure which may be generally referred to as a work unit.

The meter portal 700 may be associated with one or more users or accounts 350. Portions of the meter portal 700 may be accessible and/or modifiable based on privileges assigned to one or more users and/or accounts authorized to access the meter portal 700.

As shown, the meter portal 700 can be configured to display a total work unit hour consumption (or usage) 702, a base (or trough usage) 703, a peak usage 705, a total consumption time period 704, an average work unit hour consumption per unit of time 706, an overall (e.g., total) cost per unit of time 708, a work unit hour cost 710, a market work unit hour price 712, an idle computing system or resource tracker 714, a resource tracker 716, and a capacity projection 718. The total work unit hour consumption 702 may be represented as a number of work units consumed over the total consumption time period 704 (e.g., since the meter 118 was implemented and/or for periods of time selected by a user). As such, the average work unit hour consumption per unit of time 706 may be determined by dividing the total work unit hour consumption 702 by the total consumption time period 704 or, in some cases, by a period of time selected by a user. The peak usage 705 may be representative of maximum resource usage (e.g., as measured in work units). The base usage 703 may be representative of minimum usage (e.g., as measured in work units). For example, the resource usage may be measured over multiple measurement intervals within a time window. The peak usage 703 corresponds to the measurement interval having a usage measurement corresponding to the maximum (or peak) usage for the time window and the base usage corresponds to the measurement interval having a usage measurement corresponding to the minimum (or base) usage for the time window. In some cases, the peak usage

703 may correspond to an average of maximum (or peak) usages for the time window and the base usage 705 may correspond to an average of minimum (or base) usages for the time window. Furthermore, the peak usage 703 may correspond to an average of maximum (or peak) usages according to certain time window(s) and the base usage 705 may correspond to an average of minimum (or base) usages according to certain time window(s). The peak usage 703 may represent the average of multiple maximum usage readings for a designated time window. For example, from only the time window 5 pm to 5 am. The base usage 705 may represent the average of multiple minimum usage readings for a designated time window. For example, from only the time window 5 pm to 5 am. In these examples, the peak usage 703 would be determined by adding the maximum (or peak) usage from each respective time window measured and then divided based on the number of respective time windows being evaluated and the base usage 705 would be determined by adding the minimum (or base) usage from each respective time window measured and then divided based on the number of respective time windows being evaluated. In some cases, the meter portal 700 may be associated with one or more tags, each tag corresponding to, for example, one or more computing systems.

The overall cost per unit of time 708 may be representative of the total cost of the computing resources regardless of the quantity consumed. As such, the work unit hour cost 710 may be determined by dividing the overall cost per unit of time 708 by the average work unit hour consumption per unit of time 706. Additionally, or alternatively, the overall cost per unit of time 708 may also include hardware costs, overhead costs (e.g., data center real estate costs, electricity costs, labor costs such as managing services, security costs, networking and bandwidth costs, and/or the like), and any other costs.

The market work unit hour price 712 may be representative of a price per work unit corresponding to multiple users of one or more meters 118. For example, the market work unit hour price 712 may be representative of a price per unit of time for a work unit for a given industry, a division within a corporation (e.g., a research department), the entire market, and/or any other grouping. In other words, in some cases, the market work unit hour price 712 may also generally be described as representing a price index for all resource consuming users, one or more accounts 350, a particular industry, a particular resource supplier, a particular division within one or more corporations (e.g., research departments), and/or the like. In some cases, the market work unit hour price 712 may be based, at least in part, on an average of market costs of the resources for a plurality of users.

Additionally, or alternatively, the market work unit hour price 712 may be associated with respective resource suppliers. As such a user of the meter portal 700 can compare the market work unit hour price 712 associated with one or more resource suppliers to the work unit hour cost 710 (which is in and of itself an average work unit hour cost dependent upon the quantity of work units consumed and its relevant costs over a given period of time). This may allow a user to compare (e.g., benchmark) their costs to the prices offered by suppliers on an apples-to-apples basis. In some cases, a user may also be able to compare supplier computing systems and/or resource configurations in order to select a configuration that is suitable for various workloads executed by the user (e.g., optimize the selected computing system and/or resource configuration to a specific workload). As a result, a user may be able to analyze the work unit

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hour cost **710** to the prices offered by one or more suppliers in view of the computing system and/or resource configuration offered by a supplier.

If a different supplier can price a work unit below the work unit hour cost **710**, the user, if accessing the meter portal **700** via an account **350** having sufficient privileges, may, using an exchange **715**, purchase/acquire the computing system and/or resources, and begin executing workloads on the computing system and/or resources of the lower price supplier. Additionally, or alternatively, the exchange **715** may allow resource suppliers to place asks (or ask prices) for available computing systems and/or resources and/or allow users to place bids (or bid prices) on available computing systems and/or resources and, when an agreed upon work unit hour price and/or quantity is reached, the exchange **715** may facilitate the execution and/or clearing of the transaction. In some cases, the work unit hour price may be based per unit of time. A user may also be able to purchase/acquire computing systems and/or resources scheduled for delivery at a future date. In some cases, a user may be able to purchase/acquire rights to purchase/acquire computing systems and/or resources at a date in the future. As such, a user may be able to purchase computing systems and/or resources from a supplier at a rate below the market work unit hour price **712** associated with a particular supplier and/or index. In some cases, resources may be provided by a supplier on as-used (or as-consumed) basis. In other words, the user may not pay for a predetermined quantity of resources and/or may only pay for the quantity of work units consumed. As such, a resource supplier may sell resources on an as-consumed basis, which may allow a resource supplier to sell unused (or idle) resources during idle periods and proceed to recall those resources such that the resources can be reallocated when needed. Additionally, or alternatively, a user may purchase/acquire computing systems and/or resources from a supplier that does not have the lowest price (e.g., out of a preference for a particular supplier, location, and/or the like). In some cases, resource suppliers may price their computing systems and/or resources according to peak and off-peak hours.

On the exchange **715**, one or more work unit hours may be tokenized in the form of a smart contract that runs on a blockchain network.

The idle computing system or resource tracker **714** may be configured to represent computing systems (e.g., servers) or computing resources that are idle. An idle computing system or resource may be a computing system or resource that is only executing idle/background processes and/or is not being productive. In some cases, idle computing systems and/or resources can be identified and powered off in order to reduce electricity consumption or to avoid pay-per-use billing in a cloud environment. In some cases, the idle computing systems and/or resources can be powered off using two-way communications via the meter **118**. Furthermore, the idle computing systems and/or resources may be powered off using two-way communications via the meter **118** by way of the host operating system, guest operating system, or a hypervisor. The determination of whether a computing system or resource is idle may be based, at least in part, on a quantity of work units being consumed by a certain computing system or resource. For example, a threshold may be set that generally corresponds to the work unit hour consumption rate of the computing system or resource when no applications or operating systems are being executed thereon. As such, when the work unit hour consumption rate falls below the threshold, the corresponding computing system or resource may be identified as idle.

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In some cases, the threshold may be user adjustable (e.g., by an authorized account **350**). In this case, the threshold may not correspond to the work unit consumption of the computing system or resource when no applications or operating systems are being executed thereon. Additionally, or alternatively, idle resources may be determined based, at least in part, on one or more resource specific metrics (e.g., processor speed, memory consumption, storage consumption, graphics processor consumption, and/or the like). In some cases, when a computing system and/or resource is identified as idle, the computing system and/or resource may be offered for sale on the exchange **715** during the idle period. When an owner needs use of these resources, the resources may be recalled for use by the owner. In this scenario, an owner may be the resource supplier who owns and operates the computing resources, or it may be a user who temporarily rents computing resources from the resource supplier and then further opts to sell their idle capacity during the rental agreement while maintaining their contractual arrangement with the resource supplier. Furthermore, an owner may provide advanced notice (i.e., 30 seconds, 60 seconds, 1 hour, or any other time interval) to recall their computing resources to the user of their computing resources via the analysis platform **312**. In some cases, a user, upon receiving a recall, may be provided an option to increase their current work unit consumption rate to retain temporary ownership of the computing resources. Furthermore, the owner may have the option to decline or accept the new work unit consumption rate offered by the user.

As shown, the idle computing system or resource tracker **714** may represent idle computing systems or resources graphically (e.g., in real-time or near real-time) as a total number of idle computing systems or resources **720** and/or a total idle computing system or resource time **722**. The total idle computing system or resource time **722** may be representative of an amount of time in which at least one computing system or resource was determined to be idle. In some cases, a user may be able to assign an average cost per unit of time to run their workloads, or the meter portal itself will receive cost metrics via a plug-in(s) from a 3rd party platform that will determine the average cost per unit of time. The meter portal may also estimate an amount of potential savings by multiplying the average cost per unit of time with the total idle time (in the equivalent unit of time; i.e., minutes, hours, and/or the like) over the measurement interval. In some cases, the usage analyzer **386** may analyze historical workload behavior and idle times, and predict time(s) in the future when workloads may run idle. In this scenario, an owner may be able to sell this capacity via the exchange **715** in anticipation of it being idle. Furthermore, the owner may stipulate via their contractual agreement that they may recall resources with no advanced notice, or with advanced notice. The owner may also stipulate that they can cancel their agreement with the buyer before the buyer ever takes delivery of the computing resources; in certain cases, this may come at a cost to the owner.

The idle computing system or resource tracker **714** may be communicatively coupled to one or more of, for example, the metric/attribute collector **302** and/or the analysis platform **312**. For example, one or more of the metric/attribute collector **302** and/or the analysis platform **312** may determine an idle condition based, at least in part, on the threshold (e.g., as received from the meter portal **700**).

The resource tracker **716** may be configured to show a quantity of resources purchased **724** for a respective resource and a corresponding maximum quantity consumed **726** for a respective resource. The quantity of resources

purchased may be represented as a single value (e.g., in terms of work units) and the quantity of resources consumed may be represented as a single value (e.g., in terms of work units). Additionally, or alternatively, the quantity of resources purchased may be represented as multiple values (e.g., in terms of units corresponding to specific resources such as megahertz, gigabytes, and/or the like) and the quantity of resources consumed may be represented as multiple values (e.g., in terms of units corresponding to specific resources such as megahertz, gigabytes, and/or the like).

The capacity projection **718** may be configured to display a capacity estimate for one or more computing systems and/or resource configurations. The capacity estimate may generally correspond to a projected resource configuration(s) for meeting a user's workloads at peak utilization without encountering performance issues. The projected resource configuration may be based, at least in part, on, for example, data generated by one or more of the usage forecaster **385** and/or usage analyzer **386**. In some cases, the projected resource configuration may incorporate a resource allowance. The resource allowance may generally be described as additional resources that are in excess of what is projected and/or previously identified as the maximum usage (e.g., 5%, 10%, 15%, 25%, and/or any other percentage of resources in excess). This may allow a user to avoid not having enough resources in order to successfully run a workload without performance issues (e.g., which may sometimes be referred to as a bottleneck). The allowance may be determined automatically based, at least in part, on data received from, for example, the usage forecaster **385** and/or usage analyzer **386**. Additionally, or alternatively, the allowance may be a fixed value. For example, the fixed value may be set by a user or determined via machine learning. In some cases, the meter **118** may measure performance metrics relating to the performance of a computing system and/or resource configuration, such as TOPS (Input/Output Operations Per Second), bandwidth, memory bandwidth, and/or any other performance metric(s). Furthermore, measured performance metrics may be utilized in conjunction with physical resource usage to determine an appropriately right-sized resource configuration or computing system to run certain workload(s) via the usage forecaster **385** and/or usage analyzer **386**.

The capacity projection **718** may also be configured to display a capacity estimate for one or more graphics processing systems' and/or graphics processing resource configurations. As it relates to the graphics processing system and/or graphics processing resource configurations, the meter **118** may measure the graphics processor, graphics memory, graphics memory speed, memory bandwidth, and/or other computing resources and/or performance metrics and the usage forecaster **385** and/or usage analyzer **386** may provide a capacity estimate that generally corresponds to a projected graphics processing resource configuration(s) for meeting a user's workloads at peak utilization without encountering performance issues. In some cases, the meter **118** may measure a graphics processing system utilizing multi-instance GPU ("MIG"), which partitions a graphics processing system into multiple instances, each instance having dedicated compute, memory, and bandwidth.

In some cases, the capacity projection **718** may be configured to display a recommended computing system and/or resource configuration. The recommended computing system and/or resource configuration may be based, at least in part, on, for example, one or more of previously executed workloads, data generated by the usage forecaster **385**,

and/or data generated by the usage analyzer **386**. Additionally, or alternatively, in some cases, a recommended computing system and/or resource configuration may be based, at least in part, on a rate at which a supplier can supply resources (e.g., in terms of work unit hours supplied per unit of time). The amount and/or rate at which resources concurrently supplied may determine how quickly a workload can be completed. The recommended computing system and/or resource configurations may be representative of standardized computing system and/or resource configurations (e.g., prebuilt computing systems) and/or custom resource configurations (e.g., custom made computing systems). In some cases, a prebuilt computing system may be based upon a product inventory by a third-party resource supplier.

In some cases, the capacity projection **718** may be configured to display a recommended graphics processing system and/or graphics processing resource configuration. The recommended graphics processing system and/or graphics processing resource configuration may be based, at least in part, on, for example, one or more of previously executed workloads, data generated by the usage forecaster **385**, and/or data generated by the usage analyzer **386**. Additionally, or alternatively, in some cases, a recommended graphics processing system and/or graphics processing resource configuration may be based, at least in part, on a rate at which a supplier can supply resources (e.g., in terms of work unit hours supplied per unit of time). The amount and/or rate at which resources concurrently supplied may determine how quickly a workload can be completed. The recommended graphics processing system and/or graphics processing resource configurations may be representative of standardized computing system and/or resource configurations (e.g., prebuilt graphics processing systems) and/or custom resource configurations (e.g., custom made graphics processing systems). In some cases, a prebuilt graphics processing system may be based upon a product inventory by a third-party resource supplier.

As shown by plot **728**, the total work unit hour consumption **702** may be graphically represented over the total consumption time period **704**. For example, the total work unit hour consumption **702** may be plotted on a consumption per day basis, a consumption per hour basis, a consumption per minute basis, and/or the like.

FIG. **8** shows an example of a supply manager **800**. The supply manager **800** may include a supply calculator **802** and a cost calculator **804**. The supply calculator **802** may be configured to determine a supply rate of disparate computing resources (represented as a single unit of measure such as a work unit). The supply rate may generally be described as a quantity of work units capable of being supplied over a period of time. In other words, the supply rate indicates a rate at which disparate computing resources can be supplied as a single unit of measure over a period of time (e.g., 100 work unit hours per hour). The supply rate may be calculated experimentally (e.g., using historical data) and/or theoretically based on the computing resources being used. Additionally, or alternatively, the supply rate may be calculated based on a configuration of a computing system. In some cases, a maximum supply rate may be calculated. The supply rate may vary based on the underlying performance capability and/or work output potential of the computing resources. The maximum supply rate for a given supplier may be presented to a user through the meter portal **700** or via a 3rd party dashboard or user interface. In some cases, the supply calculator **802** may be configured to calculate an average supply rate over a period of time. Additionally, or

alternatively, the supply calculator **802** may be configured to calculate a total (or maximum) quantity of work units hours that are available to be supplied over a period of time (e.g., 100 work unit hours).

The cost calculator **804** can be configured to concurrently determine a cost to the resource supplier to supply the resources and/or to estimate a list price at which the resources may be sold in terms of a work unit hour price. The cost to supply the resources may include hardware costs, overhead costs (e.g., data center real estate costs, electricity costs, labor costs such as managing services, security costs, networking and bandwidth costs, and/or the like), and any other costs. The work unit hour price may vary based on the underlying performance capability or work output potential of the computing resources. The cost calculator **804** may use the maximum supply rate determined by the supply calculator **802** to determine a cost per work unit hour supplied. In some cases, the cost calculator **804** may be configured to factor in an average quantity of unused supply into the cost calculation. In other words, the cost calculator **804** may be configured to factor resource waste into the resource cost.

According to one aspect of the present disclosure, there is provided a computing network. The computing network may include physical computing resources configured to execute at least one workload. The computing network may also include a meter configured to measure a quantity of the physical computing resources being used over a period of time. The meter may generate a single usage value that is representative of the quantity of physical resources used over the period of time.

According to another aspect of the present disclosure, there is provided at least one computer readable storage medium having instructions stored thereon, which, when executed by one or more processors, cause the one or more processors to carry out one or more operations. The one or more operations may include measuring a quantity of one or more physical computing resources being used over a period of time. The one or more operations may also include normalizing each measured quantity of each physical computing resource being used. The one or more operations may further include summing the normalized measured quantities of the physical computing resources being used. The one or more operations may also include generating a single usage value representative of the physical computing resources being used over the period of time based on the summed normalized physical computing resources.

According to yet another aspect of the present disclosure, there is provided a method. The method may include measuring a quantity of one or more physical computing resources being used over a period of time. The method may also include normalizing each measured quantity of each physical computing resource being used. The method may further include summing the normalized measured quantities of the physical computing resources being used. The method may also include generating a single usage value representative of the physical computing resources being used over the period of time based on the summed normalized physical computing resources.

According to yet another aspect of the present disclosure, there is provided a computing network. The computing network may include a computing system configured to be communicatively coupled to a device. The computing system may be configured to execute at least one workload. The computing system may include physical computing resources configured to execute the at least one workload. The computing system may also include a meter configured

to measure a quantity of the physical computing resources being used over a period of time. The meter may generate a single usage value that is representative of the quantity of physical resources used over the period of time.

According to yet another aspect of the present disclosure, there is provided a system for metering disparate computer resources. The meter may include at least one collector. The collector may include at least one metric receiver configured to receive one or more metrics from at least one computing system over a period of time, wherein the metrics correspond to physical resource usage. The meter may also include at least one analysis platform configured to normalize each of the received metrics and sum the normalized metrics to generate a single usage value representative of the physical resource usage over the period of time.

According to yet another aspect of the present disclosure, there is provided, a computing network. The computing network may include a computing system having a plurality of physical computing resources configured to execute one or more workloads and a meter communicatively coupled to the computing system. The meter may be configured to measure a quantity of the physical computing resources being used over a period of time. The meter may include at least one collector that includes at least one metric receiver configured to receive one or more metrics from the computing system, wherein the metrics correspond to the quantity of the physical resources used. The meter may also include at least one analysis platform in communication with the collector and configured to normalize each of the received metrics and sum the normalized metrics to generate a single usage value representative of the physical resource usage over the period of time.

According to yet another aspect of the present disclosure, there is provided a computing network. The computing network may include a computing system configured to be communicatively coupled to a device. The computing system may be configured to execute at least one workload. The computing system may include physical computing resources configured to execute the at least one workload. The computing system may be configured to communicate with a meter. The meter may be configured to measure a quantity of the physical computing resources being used, wherein the meter is further configured to generate a single usage value that is representative of the quantity of physical resources used.

According to yet another aspect of the present disclosure, there is provided a computing network. The computing network may include a computing system having a plurality of physical computing resources configured to execute one or more workloads. The computing network may also include at least one metric collector having at least one metric receiver that is configured to receive one or more metrics from the computing system, wherein the one or more metrics correspond to a usage of the physical computing resources. The computing network may further include at least one backup collector having a backup receiver that is configured to receive the one or more metrics from the computing system. The computing network may also include at least one analysis platform in communication with at least one of the metric collector or backup collector, wherein the analysis platform may be configured to normalize each of the received metrics and sum the normalized metrics to generate a single usage value.

According to yet another aspect of the present disclosure, there is provided a system for metering disparate computer resources. The system may include at least one collector that includes at least one metric receiver and an account man-

ager. The metric receiver may be configured to receive one or more metrics from at least one computing system, wherein the metrics correspond to physical resource usage. The account manager may be configured to manage a plurality of accounts. Each of the plurality of accounts may be associated with at least one resource consuming user, wherein the physical resource usage of each resource consuming user is associated with a respective account. The system may also include a usage analyzer configured to generate a benchmark for one or more accounts that is representative of a per unit cost of the physical resource usage.

According to yet another aspect of the present disclosure, there is provided a computing network. The computing network may include a computing system configured to be communicatively coupled to a device. The computing system may also be configured to execute at least one workload. The computing system may include physical computing resources configured to execute the at least one workload. The computing network may also include a meter that is configured to measure a quantity of the physical computing resources being used over a period of time and generate a single usage value that is representative of the quantity of physical resources used over the period of time. The meter may be further configured to selectively disable access, by the device, to at least a portion of the computing system.

According to yet another aspect of the present disclosure there is provided a method. The method may include measuring a quantity of a plurality of physical computing resources being used over a period of time. The method may also include normalizing each measured quantity of each physical computing resource being used. The method may also include summing the normalized measured quantities of the physical computing resources being used to generate a single usage value representative of the physical computing resources being used over the period of time.

In some cases, the plurality of physical computing resources may include a plurality of disparate physical computing resources. In some cases, the plurality of physical computing resources may include one or more of a graphics processing system, a field programmable gate array, an application-specific integrated circuit, a system on chip, a digital signal processor, a microcontroller, or an adaptive compute acceleration platform. In some cases, the plurality of physical computing resources may include the graphics processing system and the graphics processing system may include a graphics processor and graphics memory. In some cases, the method may further include measuring a maximum quantity of a plurality of physical computing resources available over the period of time, normalizing each measured maximum quantity of each available physical computing resource, and summing the normalized measured maximum quantities of the available physical computing resources to generate a single available resource value representative of the maximum quantity of available physical computing resources over the period of time.

According to yet another aspect of the present disclosure there is provided a computing network. The computing network may include one or more computing systems having a plurality of physical computing resources configured to execute one or more workloads. The computing network may also include one or more metric collectors, each metric collector having at least one metric receiver that is configured to receive a plurality of metrics from the one or more computing systems. The plurality of metrics may correspond to a usage of the physical computing resources. The com-

puting network may also include one or more analysis platforms configured to normalize each of the received metrics.

In some cases, the one or more analysis platforms can be further configured to sum the normalized metrics to generate a single usage value. In some cases, at least a portion of the physical computing resource usage may be associated with one or more accounts. In some cases, the one or more accounts may include a plurality of accounts and at least one account may be associated with a portion of the physical computing resource usage and at least one other account may be associated with a different portion of the physical computing resource usage. In some cases, at least one of the metric collectors may further include a plugin database having one or more plugins, wherein the one or more plugins can be configured to enable at least one of the metric collectors to communicate with one or more computing systems. In some cases, the analysis platform can be configured to generate a notification based, at least in part, on a comparison of the physical computing resource usage to a threshold. In some cases, the one or more analysis platforms can include a plurality of analysis platforms and at least one of the analysis platforms can be a backup analysis platform. In some cases, the one or more analysis platforms can be configured to activate or deactivate at least one of the metric collectors at a predetermined time. In some cases, at least one of the metric collectors may include at least one of the one or more analysis platforms. In some cases, the one or more metric collectors may include a plurality of metric collectors and at least one of the plurality of metric collectors may be a backup metric collector. In some cases, the analysis platform may be further configured to determine one or more of a peak usage (e.g., a maximum usage), a trough usage (e.g., a minimum usage), or an average usage corresponding to each physical computing resource over the period of time based, at least in part, on the received metrics.

According to yet another aspect of the present disclosure there is provided at least one computer readable storage medium having one or more instructions stored thereon, which, when executed by one or more processors, cause one or more operations to be carried out. The operations may include measuring a quantity of a plurality of physical computing resources being used over a period of time. The operations may also include normalizing each measured quantity of each physical computing resource being used. The operations may also include summing the normalized measured quantities of the physical computing resources being used to generate a single usage value representative of the physical computing resources being used over the period of time.

In some cases, the plurality of physical computing resources may include a plurality of disparate physical computing resources. In some cases, the plurality of physical computing resources may include one or more of a graphics processing system, a field programmable gate array, an application-specific integrated circuit, a system on chip, a digital signal processor, a microcontroller, or an adaptive compute acceleration platform. In some cases, the plurality of physical computing resources may include the graphics processing system and the graphics processing system may include a graphics processor and graphics memory. In some cases, the operations may further include measuring a maximum quantity of a plurality of physical computing resources available over the period of time, normalizing each measured maximum quantity of each available physical computing resource, and summing the normalized measured maximum quantities of the available physical computing

resources to generate a single available resource value representative of the maximum quantity of available physical computing resources over the period of time. In some cases, the operations may also include accessing a plugin database, wherein the plugin database may include one or more plugins configured to enable measuring of the plurality of physical computing resources. In some cases, the operations may further include determining one or more of a peak usage (e.g., a maximum usage), a trough usage (e.g., a minimum usage), or an average usage corresponding to each physical computing resource over the period of time.

In yet another aspect of the present disclosure there is provided a method. The method may include measuring a quantity of graphics processing system resources being used over a period of time, wherein the graphics processing system resources may include a graphics processor and graphics memory. The method may also include normalizing the measured quantities corresponding to the graphics processor and graphics memory. The method may also include summing the normalized measured quantities to generate a single usage value representative of the graphics processing system resources being used over the period of time.

While several embodiments of the present disclosure have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the functions and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the present disclosure. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings of the present disclosure is/are used.

Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the disclosure described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, the disclosure may be practiced otherwise than as specifically described and claimed. The present disclosure is directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the scope of the present disclosure.

The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

The terms “couple” and “coupled,” as used herein, encompass both direct and indirect coupling unless clearly indicated to the contrary.

The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified, unless clearly indicated to the contrary.

An embodiment of the present disclosure may be implemented according to at least the following:

Clause 1: A method comprising: receiving a single usage value for a plurality of physical computing resources over a period of time; receiving a cost for the plurality of physical computing resources; and determining, using one or more processors, a normalized usage rate cost of the plurality of physical computing resources over the period of time by normalizing the cost for the plurality of physical computing resources for a unit of the single usage value.

Clause 2: A method comprising: receiving a single usage value for a plurality of computing resources over a period of time; receiving a cost for the plurality of computing resources; and determining, using one or more processors, a normalized usage rate cost of the plurality of computing resources over the period of time by normalizing the cost for the plurality of computing resources for a unit of the single usage value.

Clause 3: The method of Clauses 1 or 2 further comprising comparing the normalized cost for a unit of the single usage value with a normalized cost index for a unit of the single usage value.

Clause 4: The method of Clause 3 further comprising calculating the normalized cost index for a unit of the single usage value based on a plurality of normalized costs for a unit of the single usage value for a plurality of computing environments.

Clause 5: The method of Clause 4 further comprising normalizing the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments.

Clause 6: The method of Clauses 4 or 5 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a usage rate of respective computing environments of the plurality of computing environments.

Clause 7: The method of any of Clauses 4-6 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a total single usage value of respective computing environments of the plurality of computing environments.

Clause 8: The method of any of Clauses 4-7 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a peak usage value of respective computing environments of the plurality of computing environments.

Clause 9: The method of any of Clauses 1-8 further comprising comparing the normalized cost for a unit of the single usage value with one or more provider normalized supply charge rates from one or more computing environment providers.

Clause 10: The method of any of Clauses 3-9 further comprising calculating the normalized cost index for a unit of the single usage value based on a plurality of supply charge rates for a unit of the single usage value from a plurality of computing environment providers.

Clause 11: The method of Clause 9 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the supply charge rates from the

one or more computing environment providers based on respective maximum usage rate availability for a respective computing system.

Clause 12: The method of any of Clauses 1-11 further comprising determining a single usage value for the plurality of computing resources over the period of time.

Clause 13: The method of Clause 12, wherein determining a single usage value for the plurality of computing resources over the period of time comprises: calculating, using the one or more processors, a utilization for each of the plurality of computing resources based at least on the one or more resource attributes of each respective computing resource of the plurality of computing resources; generating, for each respective computing resource of the plurality of computing resources and based at least on a resource allocation and the resource compute portion of each respective computing resource, a conversion unit; normalizing, for each respective computing resource of the plurality of computing resources and using at least a conversion unit associated with the respective computing resource, the utilization associated with the respective computing resource; and summing normalized utilizations for each of the plurality of computing resources to generate the single usage value representative of the plurality of computing resources over the period of time.

Clause 14: The method of Clause 12, wherein determining a single usage value for the plurality of computing resources over the period of time comprises: calculating, using the one or more processors, a usage for each of the plurality of computing resources based at least on the one or more resource attributes of each respective computing resource of the plurality of computing resources over a plurality of sub-periods of the period of time; normalizing the usage for each of the plurality of computing resources for a predetermined unit of the period of time; generating, for each respective computing resource of the plurality of computing resources and based at least on a resource allocation and a resource compute portion of each respective computing resource, a conversion unit; normalizing, for each respective computing resource of the plurality of computing resources and using at least a conversion unit associated with the respective computing resource, the usage associated with the respective computing resource; and summing the normalized usages for each of the plurality of computing resources to generate the single usage value representative of the plurality of computing resources over the period of time.

Clause 15: The method of any of Clauses 1-14, wherein the cost for the plurality of computing resources is either a total cost for the plurality of computing resources over the period of time or a cost per unit of time.

Clause 16: The method of Clause 15, wherein determining the normalized usage rate cost of the plurality of computing resources over the period of time comprises: calculating, using one or more processors, a total cost for the plurality of computing resources over the period of time based on the cost per unit of time; and normalizing the total cost for the plurality of computing resources for a unit of the single usage value.

Clause 17: The method of any of Clauses 1-6, wherein determining the normalized usage rate cost of the plurality of computing resources over the period of time comprises: calculating, using one or more processors, an average single unit usage rate for a predetermined unit of time over the period of time; and normalizing the cost per unit of time for the plurality of computing resources over the period of time for the average single unit usage rate of the single usage value.

Clause 18: A method comprising: receiving a plurality of normalized costs for a unit of a single usage value for a plurality of computing environments; and determining a normalized cost index for a unit of the single usage value by combining the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments.

Clause 19: The method of Clause 18, wherein determining the normalized cost index for a unit of the single usage value comprises normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments.

Clause 20: The method of Clauses 18 or 19, wherein determining the normalized cost index for a unit of the single usage value comprises normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a usage rate of respective computing environments of the plurality of computing environments.

Clause 21: The method of any of Clauses 18-20, wherein determining the normalized cost index for a unit of the single usage value comprises normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a total single usage value of respective computing environments of the plurality of computing environments.

Clause 22: The method of any of Clauses 18-21, wherein determining the normalized cost index for a unit of the single usage value comprises normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a peak usage value of respective computing environments of the plurality of computing environments.

Clause 23: The method of any of Clauses 18-22 further comprising calculating the plurality of normalized costs for a unit of the single usage value for a plurality of computing environments based on respective single usage values and single usage value costs for each of the plurality of computing environments.

Clause 24: The method of Clause 23, wherein calculating the plurality of normalized costs for a unit of the single usage value for a plurality of computing environments comprises determining a normalized usage rate cost of a plurality of computing resources over a period of time for each of the plurality of computing environments by: calculating, using one or more processors, a total cost for the plurality of computing resources for each computing environments over the period of time; and normalizing the total cost for a unit of the single usage value for each computing environments over the period of time.

Clause 25: The method of Clause 23, wherein calculating the plurality of normalized costs for a unit of the single usage value for a plurality of computing environments comprises determining a normalized usage rate cost of a plurality of computing resources over a period of time for each of the plurality of computing environments by: calculating, using one or more processors, an average single unit usage rate for a predetermined unit of time over the period of time for each of the plurality of computing environments; and normalizing the cost per unit of time for the plurality of computing resources over the period of time for the average single unit usage rate the single usage value for the plurality of computing environments.

Clause 28: The method of any of Clauses 18-25 further comprising calculating the normalized cost index for a unit of the single usage value based on a plurality of supply

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charge rates for a unit of the single usage value from a plurality of computing environment providers.

Clause 27: The method of any of Claims 18-26 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the supply charge rates from the plurality of computing environment providers based on their maximum usage rate availability for a respective computing system.

Clause 28: The method of any of Claims 1-27, wherein the plurality of computing resources are physical computing resources comprising at least one graphics processing system, the graphics processing system having as resource attributes (i) a graphics processor variable clock speed and (ii) a variable graphics memory clock speed.

What is claimed is:

1. A method comprising:

determining, using one or more processors, a user's demand for physical computing resources based on a consumption of a plurality of physical computing resources by the user's workload;

receiving a single usage value for the plurality of physical computing resources over a period of time, the single usage value being based at least in part on a utilization for each of the plurality of physical computing resources, each utilization based at least on one or more resource attributes of each respective physical computing resource of the plurality of physical computing resources;

receiving, via a network, a cost for the plurality of physical computing resources;

determining, using one or more processors, a normalized usage rate cost of the plurality of physical computing resources over the period of time by normalizing the cost for the plurality of physical computing resources for a unit of the single usage value;

outputting, for display in a user interface configured for displaying resource usage for the plurality of physical computing resources, the normalized usage rate cost of the plurality of physical computing resources over the period of time and the workload of the plurality of physical computing resources;

outputting, for display in the user interface, a comparison of the normalized usage rate cost to a price provided by one or more resource suppliers;

determining, using one or more processors, that a price of the one or more resource suppliers is less than the normalized usage rate cost;

responsive to determining that the price of the one or more resource suppliers is less than the normalized usage rate cost, automatically requesting, by an exchange in communication with the user interface, additional physical computing resources at the price from the one or more resource suppliers;

receiving confirmation of a contract entered into between the one or more resource suppliers and the user, the receipt of the confirmation being a triggering event; automatically disabling use of at least a portion of the plurality of physical computing resources responsive to receiving the triggering event; and

utilizing the additional physical computing resources provided by the one more resource suppliers for the user's workload.

2. The method of claim 1 further comprising:

comparing the normalized cost for a unit of the single usage value with a normalized cost index for a unit of the single usage value; and

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calculating the normalized cost index for a unit of the single usage value based on a plurality of normalized costs for a unit of the single usage value for a plurality of computing environments.

3. The method of claim 2 further comprising normalizing the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments.

4. The method of claim 3, wherein normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified is based on (i) a usage rate of respective computing environments of the plurality of computing environments or (ii) a total single usage value of respective computing environments of the plurality of computing environments.

5. The method of claim 2 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing the plurality of normalized costs for a unit of the single usage value for the plurality of computing environments modified based on a peak usage value of respective computing environments of the plurality of computing environments.

6. The method of claim 1 further comprising calculating the normalized cost index for a unit of the single usage value based on a plurality of supply charge rates for a unit of the single usage value from a plurality of computing environment providers.

7. The method of claim 6 further comprising calculating the normalized cost index for a unit of the single usage value by normalizing a plurality of supply charge rates from a plurality of computing environment providers based on respective maximum usage rate availability for a respective one or more computing systems.

8. The method of claim 1 further comprising comparing the normalized cost for a unit of the single usage value with one or more computing environment provider normalized supply charge rates from one or more computing environment providers.

9. The method of claim 1 further comprising determining a single usage value for the plurality of physical computing resources over the period of time, wherein determining a single usage value for the plurality of physical computing resources over the period of time comprises:

calculating, using the one or more processors, a utilization for each of the plurality of physical computing resources based at least on the one or more resource attributes of each respective physical computing resource of the plurality of physical computing resources;

generating, for each respective physical computing resource of the plurality of physical computing resources and based at least on a resource allocation and the resource compute portion of each respective physical computing resource, a conversion unit;

normalizing, for each respective physical computing resource of the plurality of physical computing resources and using at least a conversion unit associated with the respective physical computing resource, the utilization associated with the respective physical computing resource; and

summing normalized utilizations for each of the plurality of physical computing resources to generate the single usage value being representative of the plurality of physical computing resources over the period of time.

10. The method of claim 9, wherein determining a single usage value for the plurality of physical computing resources over the period of time comprises:

calculating, using the one or more processors, a usage for each of the plurality of physical computing resources based at least on the one or more resource attributes of each respective physical computing resource of the plurality of physical computing resources over a plurality of sub-periods of the period of time;

normalizing the usage for each of the plurality of physical computing resources for a predetermined unit of the period of time;

generating, for each respective physical computing resource of the plurality of physical computing resources and based at least on a resource allocation and a resource compute portion of each respective physical computing resource, a conversion unit;

normalizing, for each respective physical computing resource of the plurality of physical computing resources and using at least a conversion unit associated with the respective physical computing resource, the usage associated with the respective physical computing resource; and

summing the normalized usages for each of the plurality of physical computing resources to generate the single usage value being representative of the plurality of physical computing resources over the period of time.

11. The method of claim 1, wherein the cost for the plurality of physical computing resources is either a total cost for the plurality of computing resources over the period of time or a cost per unit of time.

12. The method of claim 11, wherein determining the normalized usage rate cost of the plurality of physical computing resources over the period of time comprises:

calculating, using one or more processors, an average single unit usage rate for a predetermined unit of time over the period of time; and

normalizing the cost per unit of time for the plurality of physical computing resources over the period of time for the average single unit usage rate of the single usage value.

13. The method of claim 11, wherein determining the normalized usage rate cost of the plurality of physical computing resources over the period of time comprises:

calculating, using one or more processors, a total cost for the plurality of physical computing resources over the period of time based on the cost per unit of time; and normalizing the total cost for the plurality of physical computing resources for a unit of the single usage value.

14. A method comprising:

receiving a single usage value for a plurality of physical computing resources of a user over a period of time, the single usage value being based at least in part on a utilization for each of the plurality of physical computing resources, each utilization based at least on one

or more resource attributes of each respective physical computing resource of the plurality of physical computing resources;

receiving, via a network, a cost for the plurality of physical computing resources;

determining, using one or more processors, a normalized usage rate cost of the plurality of physical computing resources over the period of time by normalizing the cost for the plurality of physical computing resources for a unit of the single usage value;

determining, using one or more processors, that the normalized usage rate cost of the plurality of physical computing resources over the period of time is above a predetermined threshold;

outputting, for display in a user interface configured for displaying resource usage for the plurality of physical computing resources, the normalized usage rate cost of the plurality of physical computing resources over the period of time and an indication that a subset of the plurality of physical computing resources is idle;

determining, using one or more processors, a price for the subset of the plurality of physical computing resources;

outputting, for display in the user interface, a comparison of the price of the subset of the plurality of physical computing resources to a price provided by one or more resource suppliers;

determining, using one or more processors, that the price provided by the one or more resource suppliers is greater than the price of the subset of the plurality of physical computing resources;

responsive to determining that the price provided by the one or more resource suppliers is greater than the price of the subset of the plurality of physical computing resources, automatically selling, by an exchange in communication with the user interface, the subset of the plurality of physical computing resources at the price to the one or more resource suppliers; and

receiving confirmation of a contract entered into between the one or more resource suppliers and the user, the receipt of the confirmation being a triggering event;

automatically disabling for the user use of the subset of the plurality of physical computing resources responsive to receiving the triggering event; and

enabling access to the subset of the plurality of physical computing resources of the user to the one or more resource suppliers.

15. The method of claim 1, wherein responsive to receiving the triggering event automatically enabling access to the additional physical computing resources of the one or more resource suppliers to execute at least a portion of the user's workload.

16. The method of claim 1, wherein the additional physical computing resources on the exchange is tokenized in the form of a smart contract that runs on a blockchain network.

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